NATIONAL CHEMISTRY CONTEST CUBA 2023

ANSWERS TO THE PROBLEMS

"Every aspect of the world today – even politics and international relations – is affected by chemistry."

Linus Pauling

Ministerio de Educación
República de Cuba

FACULTAD DE QUÍMICA
UNIVERSIDAD DE LA HABANA

National Chemistry Contest 2023

Answers to the problems

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"Dear contestants, we wish we had enough gold medals to give you because, in our eyes, you are all winners; and we are certain that the effort has not been in vain, but in the future that already knocks on the door, you will be excellent professionals."

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Distribution of questions by exam day and grade

	Day 1	Day 2
10th	Problems 1, 2, 3, 4 and 5	Problems 8, 9, 10 and 11
11th	Problems 1, 2, 3 and 6	Problems 12, 13 and 14
12th	Problems 1, 3, 6 and 7	Problems 12, 15 and 16

CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's constant	$N_A = 6{,}0221 \cdot 10^{23}\ mol^{-1}$	Ideal gas law	$p \cdot V = n \cdot R \cdot T$
Gasses' constant	$R = 8{,}3145\ J \cdot K^{-1} \cdot mol^{-1}$	First Principle of Thermodynamics	$Q = \Delta U + W$
Zeroth in the Celsius's scale	$273{,}15\ K$	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485\ C \cdot mol^{-1}$	Nernst' equation	$E = E° - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{red}}{c_{ox}}$
Water ionic product at 25 °C	$K_W = 1{,}00 \cdot 10^{-14}$		$\Delta G° = -R \cdot T \cdot \ln K° = -z \cdot F \cdot \Delta E°$
Standard conditions	$p° = 100\ kPa = 1\ bar$ $c° = 1\ mol \cdot L^{-1}$	Van't Hoff i factor	$i = 1 + \alpha \cdot (\nu - 1)$
Arrhenius' equation	$k = A \cdot \exp\left(-\frac{E_A}{R \cdot T}\right)$	Cryoscopy	$\Delta T_c = i \cdot K_c \cdot b$

$1\ J = 1\ C \cdot 1\ V = 1\ kPa \cdot 1\ L = 1\ W \cdot 1\ s$ $\quad 1\ C = 1\ A \cdot 1\ s$ $\quad 1\ dm^3 = 1\ L = 1000\ mL$

1 atm = 760 Torr = 760 mmHg = 101,325 kPa $\quad 1\ eV = 96{,}485\ kJ \cdot mol^{-1}$ $\quad M \equiv mol/L$

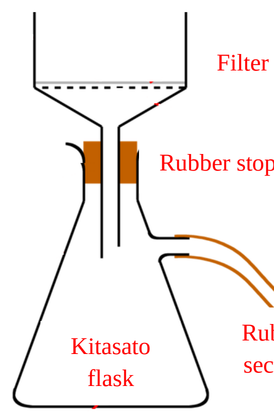
PERIODIC TABLE OF THE ELEMENTS

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1 H 1.008																	2 He 4.003
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.06	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.63	33 As 74.92	34 Se 78.97	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.95	43 Tc -	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57-71	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po -	85 At -	86 Rn -
87 Fr -	88 Ra -	89-103	104 Rf -	105 Db -	106 Sg -	107 Bh -	108 Hs -	109 Mt -	110 Ds -	111 Rg -	112 Cn -	113 Nh -	114 Fl -	115 Mc -	116 Lv -	117 Ts -	118 Og -

57 La 138.9	58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm -	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
89 Ac -	90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np -	94 Pu -	95 Am -	96 Cm -	97 Bk -	98 Cf -	99 Es -	100 Fm -	101 Md -	102 No -	103 Lr -

Problem 1. Identification of substances (common for all grades)

1.1	1.2	1.3	1.4	1.5	Total	Total
11	11	8	3	7	40	10%
11	11	8	3	7	40	10%



A concentrated aqueous solution (problem solution) of a blue colored salt **1** was prepared. Using pH paper, it was determined that the solution was practically neutral (neither acid nor basic). The coloration of the solution on the flame was blue-green. Next, a few milliliters of the problem solution were taken and placed in several test tubes, arranged in a rack. A concentrated sodium hydroxide solution was added to the first test tube and the appearance of a "sky-blue" (light blue) precipitate was observed. When excess sodium hydroxide was added, the precipitate dissolved, resulting in a deep blue solution. In the second test tube, excess concentrated hydrochloric acid was added and the solution resulting from the mixture had a green-yellow color. A solution of concentrated barium chloride was added to the third test tube and the formation of a white precipitate was observed. A solution of concentrated calcium chloride was added to the fourth test tube and the formation of a white precipitate was observed.

1.1. Indicate the cation and anion present in salt **1**. Write all the equations (ionic or molecular) of the chemical reactions that allowed their identification.

Cu^{2+} **3 mark** SO_4^{2-} **3 mark**

$Cu^{2+}(ac) + 2OH^-(ac) \rightarrow Cu(OH)_2(s)$ **1 mark**

$Cu(OH)_2(s) + 2OH^-(ac) \rightarrow [Cu(OH)_4]^{2-}(s)$ **1 mark**

It is acceptable to represent the tetrahydroxidecuprate(II) ion as CuO_2^{2-}.

$Cu^{2+}(ac) + 4Cl^-(ac) \rightarrow [CuCl_4]^{2-}(s)$ **1 mark**

$Ba^{2+}(ac) + SO_4^{2-}(ac) \rightarrow BaSO_4(s)$ **1 mark**

$Ca^{2+}(ac) + SO_4^{2-}(ac) \rightarrow CaSO_4(s)$ **1 mark**

Finally, a portion of the test solution was concentrated by heating; subsequently, it was cooled and blue crystals of salt **1** were obtained. The solid was separated by vacuum filtration, placed in a porcelain dish, on top of a tripod and heated slightly with a Bunsen burner. The capsule was held with pincers and carefully stirred with a glass shaker until all the solid was observed to turn into a greyish-white powder.

1.2. About the experimental procedures, answer:

1.2.a) What does the change on the color of the solid during heating indicate?

A chemical reaction occurs. / Loss of water of hydration occurs. / The substance decomposed.

1 mark

1.2.b) Indicate in the vacuum filtration apparatus represented in the figure: Büchner funnel, Kitasato collecting flask, rubber section, filter paper and perforated rubber stopper.

5 mark in total (1 mark for identifying each tool in the figure).

1.2.c) State the function of the tools: rack, test tube and spoon spatula.

The rack is used to hold test tubes. **1 mark**

The test tube is used to carry out chemical reactions. **1 mark**

The spatula is used to transfer samples of solids. **1 mark**

1.2.d) Mention four protection tools used in the chemistry laboratory.

Glasses, gloves, mask, gown. **2 mark**

The concentrated aqueous solution of a colorless salt **2** was subjected to an analysis similar to that described above. When adding hydrobromic acid, a light-yellow precipitate was obtained, which dissolved in concentrated ammonia and no gas evolution was observed. The addition of sodium sulfide caused a black precipitate to form and an increase in the acidity of the solution. When the pH of the original solution was increased by adding sodium hydroxide, the appearance of a brown precipitate was observed, which dissolved in concentrated ammonia. It was also observed that the solution of salt **2** turned dark colored while time passed.

1.3. Indicate the cation present in salt **2**. Write all the equations (ionic or molecular) of the chemical reactions that allowed their identification.

Ag^+ **3 mark**

$Ag^+(ac) + Br^-(ac) \rightarrow AgBr(s)$ **1 mark**

$AgBr(s) + 2NH_3(ac) \rightarrow [Ag(NH_3)_2]Br(ac)$ **1 mark**

$2Ag^+(ac) + S^{2-}(ac) \rightarrow Ag_2S(s)$ **1 mark**

$2Ag^+(ac) + 2OH^-(ac) \rightarrow Ag_2O(s) + H_2O(l)$ **1 mark**

$Ag_2O(s) + H_2O(l) + 4NH_3(ac) \rightarrow 2[Ag(NH_3)_2]OH(ac)$ **1 mark**

It is acceptable to write AgOH instead of silver oxide.

1.4. About the experimental procedures, answer:

1.4.a) Why do solutions of **2** turn black while they age?

Silver compounds are photosensitive. **1 mark**

1.4.b) Which of the following anions could be present in salt? Cl^-, NO_3^-, HCO_3^-, CO_3^{2-}, $CH_3CO_2^-$, SO_4^{2-}, $Cr_2O_7^{2-}$, I^-, SO_3^{2-}, PO_4^{3-}, $S_2O_3^{2-}$, $C_2O_4^{2-}$.

NO_3^- **1 mark** $S_2O_3^{2-}$ **1 mark**

It is also acceptable to write the ions $CH_3CO_2^-$ and SO_4^{2-} whose silver salts are sparingly soluble.

The yellow, concentrated aqueous solution of a salt **3** changed to an orange coloration when acidified with concentrated sulfuric acid. More acid was added to a portion of the orange solution until a deep red needle-shaped precipitate was obtained, the product of a non-redox reaction. Hydrogen peroxide was added to a portion of the orange solution, the solution turned green after some time and released a colorless, odorless gas. Finally, when the solutions of salts **2** and **3** were mixed, an intense red precipitate was obtained.

1.5. Indicate the anion present in salt **3**. Write all the equations (ionic or molecular) of the reactions that allowed their identification.

CrO_4^{2-} **3 mark**

$2CrO_4^{2-}(ac) + 2H^+(ac) \rightarrow Cr_2O_7^{2-}(s) + H_2O(l)$ **1 mark**

$Cr_2O_7^{2-}(s) + 2H^+(ac) \rightarrow 2CrO_3(s) + H_2O(l)$ **1 mark**

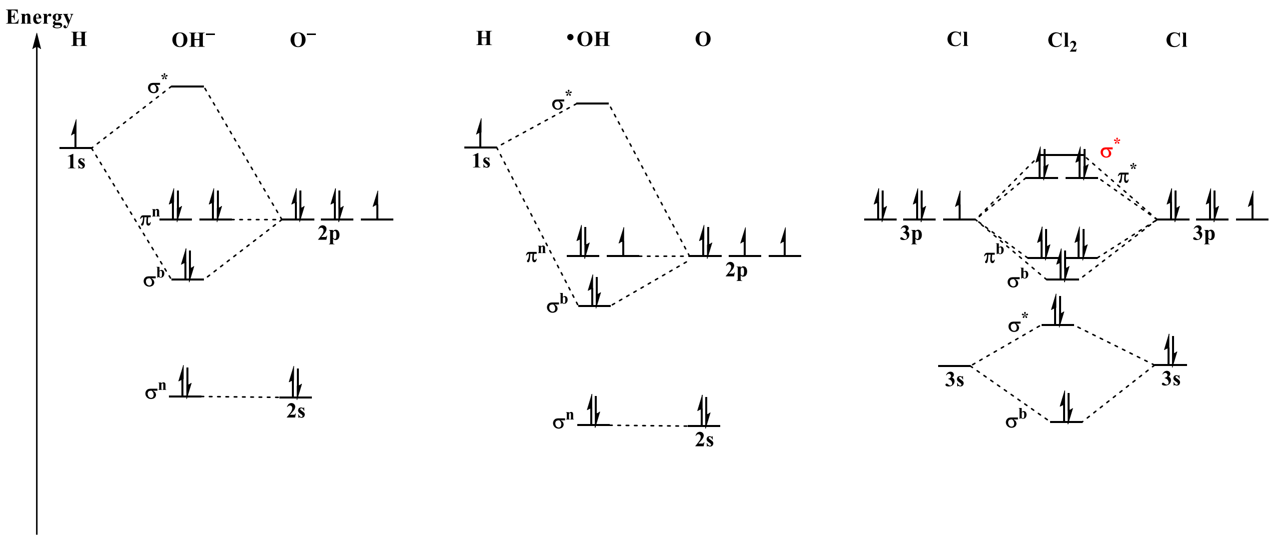
$Cr_2O_7^{2-}(s) + 3H_2O_2(l) + 8H^+(ac) \rightarrow 3O_2(g) + 2Cr^{3+}(ac) + 7H_2O(l)$ **1 mark**

$2Ag^+(ac) + CrO_4^{2-}(ac) \rightarrow Ag_2CrO_4(s)$ **1 mark**

Problem 2. Reactivity (only 10th and 11th grades)

2.1	2.2	2.3	Total	Total
24	7	9	40	10%
24	7	9	40	10%

The figure illustrates the molecular orbitals (MO) diagrams for the hydroxide anion, the hydroxyl radical and the chlorine molecule. In addition, you have additional information.



Transition state (‡): Particular, unstable configuration of the atoms during a reaction that has the maximum energy. When the atoms of the reactants "abandon" this intermediate state in the reaction path (coordinate), the formation of products will always occur. The activation energy is the difference between the energy of the transition state and the energy of the reactants.

MO: Molecular orbital, AO: atomic orbital, FO: frontier orbital, HOMO: Highest energy occupied molecular orbital, LUMO: Lowest energy unoccupied molecular orbital, SOMO: Molecular orbital occupied by a single electron, electrophile or Lewis acid: electron density deficient species, nucleophile or Lewis base: electron density rich species.

Note: In the original syllabus, an **error** appears in the MO diagram, the MO σ^* appears as σ^b, so the wrong answers that are a consequence of this mistake were accepted.

2.1. Complete the following statements as briefly as possible.

2.1.a) The O atom attracts with __more__ strength the valence electrons when compared to the Cl atom. Therefore, the energy of the OAs of O is lower than the energy of the OAs of Cl and, consequently, the O atom is _more___ electronegative than the Cl atom.

2.1.b) Species OH^- and Cl_2 are _expelled from_ while radical OH is _attracted to_ an intense magnetic field, because the first ones have all their electrons _paired_, while the last one _has unpaired electrons_.

2.1.c) The bond order in all three species is equal to _one_.

2.1.d) In Cl_2 the magnitude of the interaction (overlap) between the AOs is _smaller_ in comparison with the species OH^- and ·OH. This is because the atomic radius of the chlorine atom is _larger_.

2.1.e) The energy of the MOs in the anion OH^- is _larger/more positive/less negative_ than the energy of the MOs in the radical ·OH because _it has a negative charge_.

2.1.f) The frontier orbitals of these species are: HOMO π^n and _LUMO σ^*_ for the anion OH^-, _SOMO π^n _ and LUMO σ^* for the radical ·OH, and _LUMO σ^*_ and _HOMO π^*_ for the Cl_2 molecule.

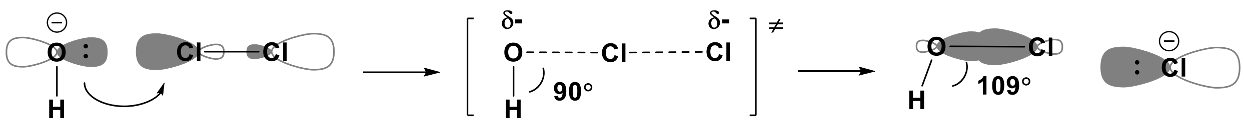
2.1.g) Partial charge takes a positive, zero, and negative value on the atoms _hydrogen_, _chlorine_ and _oxygen_, respectively.

2.1.h) Anion OH^- in its reactions behaves as _nucleophile/electron donor/base_ because it has a high electron density. On the contrary, radical ·OH and Cl_2 molecule behave as _electrophiles/electron acceptors/acids_.

2.1.i) Anion OH^- and radical ·OH are very reactive species because the first one has _negative charge_, while the second one _possesses one unpair electron_.

2.1.j) In all cases, the number of MOs that are formed are numerically _equal_ to the total number of AOs of the atoms originating the species.

The reaction mechanism between the OH^- anion and the Cl_2 molecule is represented in the following scheme, where the FOs involved in the reaction, the direction of approach of the reactants, the geometries of the transition state and of the final product and the flow of electrons are highlighted.



2.2. Complete the following statements as briefly as possible.

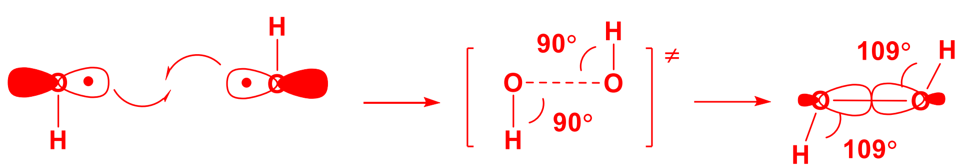
2.2.a) Anion OH^- donates the electron pair occupying the filled HOMO π^n to the empty _LUMO σ*_ of Cl_2 molecule.

2.2.b) O-H bond _is not affected/doesn't break_ during the reaction because the FO of OH^- that participates is non-bonding, while the Cl-Cl bond _is broken_ during the reaction because the FO of Cl_2 that participates is _antibonding_.

2.2.c) When an additional OH^- ion reacts with hypochlorous acid, anion _hypochlorite_ and a neutral molecule of _water_ are obtained. During this reaction, OH^- _donates_ the electron pair occupying the filled HOMO π^n to the empty LUMO σ*($2sp^3$O–1sH) of HClO.

1 mark is given per each correct statement.

2.3. Draw a scheme similar to the one shown above for the reaction that occurs between two ·OH radicals forming hydrogen peroxide. Represent the FOs involved in the reaction, the direction of approach of the reactants, the geometries of the transition state and the final product, and the electron flow (by arrows).



1 mark **1 mark** **1 mark** **2 mark** **1 mark** **2 mark** **1 mark**

Problem 3. Gold alloys (common for all grades)

3.1.	3.2	3.3	3.4	Total	Total
6	9	16	9	40	10%
6	9	16	9	40	10%

Gold is a non-reactive metal used since ancient times in coins, jewelry and as a symbol of power. Its metallic luster is characteristic and is not lost in the presence of other metals. For this reason, it is common that those who trade gold try to deceive their clients with gold alloys with different metals. The jeweler classifies the samples that contain gold in carats (22, 18 or 12 K) depending on the mass content of gold present (equal to or greater than 91,7%, 75% or 50%, respectively).

A laboratory worker was given 100 mg of different samples containing gold for analysis, obtaining the results below. All the experiments were carried out at 0 °C and 101.325 kPa.

Calculate the number of carats (K) each gold sample contains. Formulate the balanced equations of all the chemical reactions that occur and find what other metal forms the pink gold alloy.

3.1. Sample of purple gold, containing aluminum: When dissolved in excess hydrochloric acid it generates 30.0 mL of a colorless gas. V_m = 22,4 L/mol.

$Al(s) + 3H^+(ac) \rightarrow Al^{3+}(ac) + \frac{3}{2}H_2(g)$ **1 mark**

$n(H_2) = \frac{v(H_2)}{V_m} = \frac{30,0}{22,4} = 1,34\ \text{mmol}$ **1 mark**

$n(Al) = \frac{2}{3} \cdot n(H_2) = \frac{2}{3} \cdot 1,34 = 0,893\ \text{mmol}$ **1 mark**

$m(Al) = A_r(Al) \cdot n(Al) = 26,98 \cdot 0,893 = 24,09\ \text{mg}$ **1 mark**

$m(Au) = 100 - m(Al) = 100 - 24,09 = 76\%$ **1 mark** → 18 K gold **1 mark**

3.2. Sample of pink gold, which contains a metal **X**: It was dissolved stoichiometrically in 36.5 mL of a solution of nitric acid (M = 63,02 g/mol) of mass concentration 5,00 g/L, leaving a dry residue of 54, 0 mg unreacted.

$m(Au) = 54\ \text{mg} = 54\%$ **1 mark** → 12 K gold **1 mark**

$M(s) + x\,HNO_3(ac) \rightarrow$

$m(M) = 100 - 54 = 46\ \text{mg}$ **1 mark**

$m(HNO_3) = \rho(HNO_3) \cdot V(HNO_3) = 5,00 \cdot 36,5 = 182,5\ \text{mg}$ **1 mark**

$n(HNO_3) = \frac{m(HNO_3)}{M(HNO_3)} = \frac{182,5}{63,02} = 2,90\ \text{mmol}$ **1 mark**

$n(M) = \frac{n(HNO_3)}{x}$ **1 mark** → $A_r(M) = \frac{m(M)}{n(M)} = \frac{46,0}{2,90} \cdot x = 15,86 \cdot x$ **1 mark**

$x = 1 \quad A_r(M) = 15,86\ g \cdot mol^{-1}$ O?

$x = 2 \quad A_r(M) = 31,72\ g \cdot mol^{-1}$ S?

$x = 3 \quad A_r(M) = 47,59\ g \cdot mol^{-1}$ Ti?

Titanium(IV) is formed under oxidizing conditions. However, this response is acceptable and the error is carried over.

$x = 4 \quad A_r(M) = 63,45\ g \cdot mol^{-1}$ Cu **1 mark**

$Cu(s) + 4HNO_3(ac) \rightarrow Cu(NO_3)_2(ac) + 2NO_2(g) + 2H_2O(l)$ **1 mark**

3.3. Sample of white gold, which contains platinum: It was dissolved in excess of aqua regia (HNO_3/HCl) until completely dissolved, generating 35,0 mL of reddish-brown NO_2. It is known that when dissolving the sample two soluble products are formed whose percentage compositions by mass are Au : Cl : H equal to 58,0% : 41,7% : 0,3% and Pt : Cl : H equal to 47,6% : 51,9% : 0,5%. Find the formulas of the species formed.

Assuming 100 g of the gold product

$n(Au) = \frac{m(Au)}{A_r(Au)} = \frac{58,0}{196,97} = 0,294\ mol$ **1 mark**

$n(Cl) = \frac{m(Cl)}{A_r(Cl)} = \frac{41,7}{35,45} = 1,176\ mol$ **1 mark**

$n(H) = \frac{m(H)}{A_r(H)} = \frac{0,3}{1,01} = 0,30\ mol$ **1 mark**

$n(Au) : n(Cl) : n(H) = 0,294 : 1,176 : 0,30 \approx 1 : 4 : 1 \quad \rightarrow \quad H[AuCl_4]$ **1 mark**

$Au(s) + 3HNO_3(ac) + 4HCl(ac) \rightarrow H[AuCl_4](ac) + 3NO_2(g) + 3H_2O(l)$ **1 mark**

Assuming 100 g of the platinum product

$n(Pt) = \frac{m(Pt)}{A_r(Pt)} = \frac{47,6}{195,97} = 0,243\ mol$ **1 mark**

$n(Cl) = \frac{m(Cl)}{A_r(Cl)} = \frac{51,9}{35,45} = 1,46\ mol$ **1 mark**

$n(H) = \frac{m(H)}{A_r(H)} = \frac{0,5}{1,01} = 0,50\ mol$ **1 mark**

$n(Au) : n(Cl) : n(H) = 0,243 : 1,46 : 0,50 \approx 1 : 6 : 2 \quad \rightarrow \quad H_2[PtCl_6]$ **1 mark**

$Pt(s) + 4HNO_3(ac) + 6HCl(ac) \rightarrow H_2[PtCl_6](ac) + 4NO_2(g) + 4H_2O(l)$ **1 mark**

$n(NO_2) = \frac{V(NO_2)}{V_m} = \frac{35,0}{22,4} = 1,56\ mmol$ **1 mark**

$1,56 = 3 \cdot n(Au) + 4 \cdot n(Pt) = 3 \cdot \frac{m(Au)}{A_r(Au)} + 4 \cdot \frac{m(Pt)}{A_r(Pt)} \quad (1)$ **1 mark**

$100\ mg = m(Au) + m(Pt) \quad (2)$ **1 mark**

Solving the system of equations (1) and (2), we obtain

$m(Pt) = 7,00\ mg \quad \rightarrow \quad m(Au) = 93,0\ mg = 93\%$ **2 mark** $\rightarrow$ 22 K gold **1 mark**

3.4. Sample of blue gold, which contains iron: The sample was immersed in 25,0 g of a 4,00% mass silver nitrate solution until the iron completely reacted, which oxidized to the 2+ state. The treated gold sample was dried and weighed, resulting in a mass of 172 mg. The remaining silver nitrate solution was reacted with excess sodium chloride. The silver chloride solid after drying had a mass of 714 mg.

Answer 1

$Fe(s) + 2AgNO_3(ac) \rightarrow 2Ag(s) + Fe(NO_3)_2(ac)$ **1 mark**

$m(Fe) + m(Au) = 100\ mg$ **1 mark** (1)

$m(Ag)_{reac} + m(Au) = 172\ mg$ **1 mark**

$m(Ag)_{reac} = A_r(Ag) \cdot n(Ag)_{reac} = A_r(Ag) \cdot 2 \cdot \frac{m(Fe)}{A_r(Fe)}$ **2 mark**

$\frac{107,87 \cdot 2}{55,85} \cdot m(Fe) + m(Au) = 172\ mg \quad (2)$ **1 mark**

Solving the system of equations (1) and (2), we obtain

$m(Fe) = 25\ mg \quad \rightarrow \quad m(Au) = 75\ mg = 75\%$ **2 mark** 18 K gold **1 mark**

Answer 2

$Fe(s) + 2AgNO_3(ac) \rightarrow 2Ag(s) + Fe(NO_3)_2(ac)$ **1 mark**

$2 \cdot n(Fe) = n(AgNO_3)_{reac} = n(AgNO_3)_{total} - n(AgNO_3)_{exc}$ **1 mark**

$M(AgNO_3) = A_r(Ag) + A_r(N) + 3 \cdot A_r(O) = 107{,}87 + 14{,}01 + 3 \cdot 16{,}00 = 169{,}88\ g \cdot mol^{-1}$ **1 mark**

$M(AgCl) = A_r(Ag) + A_r(Cl) = 107{,}87 + 35{,}45 = 143{,}32\ g \cdot mol^{-1}$ **1 mark**

$m(AgNO_3)_{total} = \frac{4{,}00}{100} \cdot 25{,}0 = 1{,}00\ g \quad \rightarrow \quad n(AgNO_3)_{total} = \frac{1{,}00}{169{,}88} = 0{,}00589\ mol$ **1 mark**

$n(AgNO_3)_{exc} = n(AgCl) = \frac{m(AgCl)}{M(AgCl)} = \frac{0{,}714}{143{,}32} = 0{,}00498\ mmol$ **1 mark**

$n(AgNO_3)_{reac} = 0{,}00589 - 0{,}00498 = 0{,}00091\ mol$ **1 mark**

$n(Fe) = \frac{1}{2} \cdot n(AgNO_3)_{reac} = \frac{1}{2} \cdot 0{,}00091 = 0{,}000455\ mol$

$m(Fe) = A_r(Fe) \cdot n(Fe) = 55{,}85 \cdot 0{,}000455 = 25\ mg$ **1 mark**

$m(Au) = 100 - 25 = 75\ mg = 75\%$ 1 mark 18 K gold **1 mark**

Problem 4. Balancing equations (10th grade)

4.1	…	Total	Total
4	idem	40	10%
4	idem	40	10%

Complete and balance the following chemical equations.

4.1. $K(s) + O_2(g, excess) \rightarrow KO_2(s)$ **4 mark**

The product obtained is unstable and decomposes on heating.

Penalize with -1 mark if K_2O_2 is written as product.

4.2. $CaO(l) + 3C(s) \xrightarrow{\Delta} CaC_2(s) + CO(g)$ **4 mark**

CO_2 is acceptable as the product instead of CO. Penalize with -2 mark if Ca is written as the product.

4.3. $2LiNO_3(s) \xrightarrow{\Delta} Li_2O(s) + 2NO_2(g) + \frac{1}{2}O_2(g)$ **4 mark**

A reddish-brown toxic gas and a very stable white solid are obtained.

The decomposition product is temperature dependent and can could be nitrite or lithium peroxide. However, these products are unstable and discarded with the data offered and are penalized with -2 mark. It is penalized with -1 mark if NO is written as the product.

4.4. $Cu(s) + 4HNO_3(conc) \xrightarrow{\Delta} Cu(NO_3)_2(ac) + 2NO_2(g) + 2H_2O(l)$ **4 mark**

4.5. $Si(s) + 2F_2(g) \xrightarrow{\Delta} SiF_4(g)$ **4 mark**

Penalize with -2 mark if SiF_2 is written as the product.

4.6. $GeCl_4(g) + Ge(s) \xrightarrow{\Delta} 2GeCl_2(s)$ **4 mark**

It is a redox reaction.

4.7. $PCl_3(l) + 3H_2O(l) \xrightarrow{\Delta} H_3PO_3(l) + 3HCl(g)$ **4 mark**

4.8. $SO_2(g) + \frac{1}{2}O_2(g) + H_2O(l) \xrightarrow{\Delta} H_2SO_4(l)$ **4 mark**

This reaction is used in industry and requires a catalyst.

4.9. $F_2(g) + O_2(g) \rightarrow O_2F_2(g)$ **4 mark**

In the product, oxygen has an oxidation number of 1+.

It is penalized with -1 mark if OF is written as the product.

4.10. $2NaOH(conc) + SiO_2(s) \xrightarrow{\Delta} Na_2SiO_3(s) + H_2O(g)$ **4 mark**

Formulas $Na_2[SiO_2(OH)_2]$ and Na_4SiO_4 are also acceptable.

It is penalized with -1 mark for any unbalanced equation.

Problem 5. Miscellanea (10th grade)

5.1	…	Total	Total
4	idem	40	10%
4	idem	40	10%

Select the most appropriate statement in each case. You do not need to justify your answer.

5.1. They form a yellow precipitate when reacting:
A) $AgNO_3$ and K_2S B) KI and $Pb(NO_3)_2$ C) $CaCl_2$ and NaOH D) $FeSO_4$ and KOH

5.2. Which compound has the smallest percent by mass of the metal?
A) $CaCl_2$ B) $MnCl_2$ C) $FeCl_2$ D) $ZnCl_2$

5.3. From species: I) SF_2, II) SF_4 and III) SF_6 are polar:
A) I, II and III B) only I C) only III D) only II and III E) only I and II

5.4. The substance most soluble in water:
A) H_2S B) $Ca(OH)_2$ C) O_2 D) C_2H_5OH E) SiO_2

5.5. "Noble" metal, not very reactive.
A) Mn B) Rb C) Zn D) Pt

5.6. The first three ionization energies of element X are: 590; 1145 and 4912 kJ/mol. The formula for the most probable and stable ion of X is:
A) X^{1+} B) X^{2+} C) X^{3+} D) X^{1-}

5.7. Considering the following chemical equations and their ΔHº. Select the **incorrect** statement.
$2\ H_2(g) + 2\ Cl_2(g) = 4\ HCl(g)$ ΔHº = -92,3 kJ.
A) ΔHº of the inverse reaction is +92,3 kJ.
B) Four H-Cl bonds are stronger than four H-H and Cl-Cl bonds (two of each kind).
C) If HCl is produced in liquid form then ΔHº = -92,3 kJ.
D) When 1 mole of HCl(g) is formed, about 23,1 kJ of energy is released as heat.

5.8. The enthalpy change is equal to the internal energy change in the equation:
A) $C\ (s) + O_2\ (g) = CO_2\ (g)$ B) $N_2\ (g) + 2\ O_2\ (g) = 2\ NO_2\ (g)$
C) $2\ H_2O_2\ (l) = 2\ H_2O\ (l) + O_2\ (g)$ D) $C_2H_4\ (g) + 3\ O_2\ (g) = 2\ CO_2\ (g) + 2\ H_2O\ (l)$

5.9. Element with 7 electrons in "s" type orbitals.
A) Na B) N C) K D) C

5.10. If the following bond distances in pm are known: O-O: 148; Mg-O: 215; H-S: 133; S-S: 204. The bond distance of Mg-H in pm is:
A) 182 B) 202 C) 172 D) 348

4 marks are awarded for each correct answer.

Problem 6. Alcohol oxidation (only 11th and 12th grade)

6.1	6.2	6.3	6.4	6.5	6.6	6.7	6.8	6.9	6.10	6.11	Total	Total
4	18	6	4	4	5	5	13	10	6	5	80	20%
4	18	6	4	4	4	6	13	10	6	5	80	20%

Organic alcohols can react the dichromate ion $Cr_2O_7^{2-}$ in aqueous solution in an acid medium according to the following reaction. The color change experienced by the solution allows the study of this reaction by absorption spectroscopy and its use in breath analyzers.

$$Cr_2O_7^{2-}(ac) + 3RCH_2OH(ac) + 8H_3O^+(ac) \rightarrow 3RCHO(ac) + 2Cr^{3+}(ac) + 15H_2O(l)$$

6.1. Write the oxidation and reduction half equations for the reaction above.

HER: $Cr_2O_7^{2-}(ac) + 14H^+(ac) + 6e^- \rightarrow 2Cr^{3+}(ac) + 7H_2O(l)$ **2 mark**

HEO: $RCH_2OH(ac) \rightarrow RCHO(ac) + 2H^+(ac) + 2e^-$ **2 mark**

In a laboratory, 0,10 M ethanol and 0,75 M potassium dichromate solutions at pH 3,5 are available. To determine the kinetic parameters of the previous reaction, three 100 mL volumetric flasks were prepared by adding a different volume of the ethanol solution in each one. Then they were completed up to the mark with the potassium dichromate solution until three solutions of 100 mL were obtained in each case. A beam of light was passed through a spectrophotometer at a wavelength where only dichromate absorbs, and the absorbance was measured at the beginning of the mixture and 20 s later. All experiments were carried out at 25 °C. Using the molar absorptivity of dichromate, it was possible to measure the initial speed of the reaction, obtaining the data that is summarized in the following table:

Volumetric flask 100 mL	Volume of RCH_2OH (mL)	Initial rate (M/s)
1	50,0	$2{,}401 \cdot 10^{-9}$
2	60,0	$2{,}577 \cdot 10^{-9}$
3	70,0	$2{,}604 \cdot 10^{-9}$

Assume that after forming the three solutions the pH is approximately 4.5. The rate law of the reaction has the general formula $v_r = k_r \cdot [RCH_2OH]^x\ [Cr_2O_7^{2-}]^y\ [H_3O^+]^z$, where z = 1.

6.2. Calculate the partial orders of reaction (x, y) and the rate constant using the method of initial rates and the data in the table. To do this, calculate the concentrations of alcohol and dichromate in each flask after the solutions have been mixed.

Note: If you cannot solve the previous part, consider that k_r = 0,00250 u.

From the initial volumes and concentrations, the concentrations in each flask can be calculated.

Flask	V(alcohol)	c(alcohol) mol/L		$V(Cr_2O_7^{2-})$	$c(Cr_2O_7^{2-})$ mol/L		v_0 mol/(L·s)
1	50	0,05	**1 mark**	50	0,375	**1 mark**	$2{,}401 \cdot 10^{-9}$
2	60	0,06	**1 mark**	40	0,300	**1 mark**	$2{,}577 \cdot 10^{-9}$
3	70	0,07	**1 mark**	30	0,225	**1 mark**	$2{,}604 \cdot 10^{-9}$

$$v = k \cdot c^x(RCH_2OH) \cdot c^y\left(Cr_2O_7^{2-}\right) \cdot c(H^+)$$

$$c(H^+) = 10^{-4,5} = 3{,}16 \cdot 10^{-5}\ mol \cdot L^{-1}$$

$$\ln(2{,}401 \cdot 10^{-9}) = \ln k + x \cdot \ln 0{,}05 + y \cdot \ln 0{,}375 + \ln(3{,}16 \cdot 10^{-5}) \quad (1)$$

$$\ln(2{,}577 \cdot 10^{-9}) = \ln k + x \cdot \ln 0{,}06 + y \cdot \ln 0{,}300 + \ln(3{,}16 \cdot 10^{-5}) \quad (2)$$

$\ln(2{,}604 \cdot 10^{-9}) = \ln k + x \cdot \ln 0{,}07 + y \cdot \ln 0{,}225 + \ln(3{,}16 \cdot 10^{-5})$ (3)

Solving the system of equations (1), (2) and (3), we obtain

$\ln k = -6{,}00$ **3 mark** $x = 0{,}998 \approx 1$ **3 mark** $y = 0{,}499 \approx 0{,}5$ **3 mark**

$k = e^{-6} = 2{,}48 \cdot 10^{-3}\ L^{3/2} \cdot mol^{-3/2} \cdot s^{-1}$ **3 mark**

Penalize with -1 mark if the units of the constant are incorrect.

This reaction has been widely studied and it is known that it occurs through a mechanism that consists of six steps as shown below:

[1] $Cr_2O_7^{2-} + H_2O \underset{k-1}{\overset{k1}{\rightleftharpoons}} 2HCrO_4^-$

[2] $HCrO_4^- + H_3O^+ \underset{k-2}{\overset{k2}{\rightleftharpoons}} H_2CrO_4 + H_2O$

[3] $H_2CrO_4 + RCH_2OH \underset{k-3}{\overset{k3}{\rightleftharpoons}} RCH_2OCrO_3H + H_2O$

[4] $RCH_2OCrO_3H \xrightarrow{k4} RCHO + Cr(IV)$

[5] $Cr(IV) + Cr(VI) \xrightarrow{k5} 2Cr(V)$

[6] $2Cr(V) + 2RCH_2OH \xrightarrow{k6} 2Cr^{3+} + 2RCHO$

6.3. If equation **[4]** is the slow step and equilibria **[1]**, **[2]**, and **[3]** are fast, find an expression for the rate law assuming that these equilibria hold during the reaction. Express the total rate constant in terms of the rate constants for each steps.

$v = k_4 \cdot [RCH_2OCrO_3H]$ **1 mark**

$v = k_4 \cdot \frac{k_3}{k_{-3}} \cdot [RCH_2OH][H_2CrO_4]$ **1 mark**

$v = k_4 \cdot \frac{k_3}{k_{-3}} \cdot [RCH_2OH] \cdot \frac{k_2}{k_{-2}} \cdot [H_3O^+][HCrO_4^-]$ **1 mark**

$v = k_4 \cdot \frac{k_3}{k_{-3}} \cdot [RCH_2OH] \cdot \frac{k_2}{k_{-2}} \cdot [H_3O^+] \cdot \sqrt{\frac{k_1}{k_{-1}} \cdot [Cr_2O_7^{2-}]}$ **2 mark**

$v = k' \cdot [RCH_2OH] \cdot [H_3O^+] \cdot [Cr_2O_7^{2-}]^{\frac{1}{2}} \rightarrow k' = k_4 \cdot \frac{k_3}{k_{-3}} \cdot \frac{k_2}{k_{-2}} \cdot \sqrt{\frac{k_1}{k_{-1}}}$ **1 mark**

It is acceptable to find a more complex expression by applying the steady-state theory.

6.4. The reaction under study follows the Arrhenius equation. Briefly explain what are the terms activation energy and pre-exponential factor that appear in this equation.

Activation energy: the minimum energy required for the reactants to transform into the products. **2 mark**

Pre-exponential factor: is the product of the frequency factor and the steric factor, which are related to the number of collisions and direction in which occur, respectively. **2 mark**

6.5. When heating the previous solutions up to 33 ºC, the rate constant of the experimentally determined reaction increased by 2,78 times. Calculate the activation energy and the pre-exponential factor.

$$\ln\left(\frac{k_2}{k_1}\right) = -\frac{E_A}{R}\left(\frac{1}{T_2}-\frac{1}{T_1}\right)$$

$$E_A = \frac{-R\cdot\ln\left(\frac{k_2}{k_1}\right)}{\frac{1}{T_2}-\frac{1}{T_1}} = \frac{-8{,}3145\cdot\ln 2{,}78}{\frac{1}{306{,}15}-\frac{1}{298{,}15}} = 96991\ \mathrm{J\cdot mol^{-1}} = 97{,}0\ \mathrm{kJ\cdot mol^{-1}}$$ **2 mark**

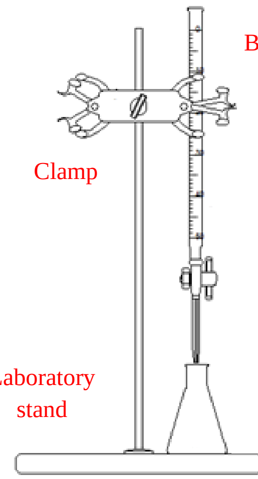
$$k = A\cdot e^{-\frac{E_A}{R\cdot T}} \quad\rightarrow\quad 2{,}48\cdot 10^{-3} = A\cdot e^{-\frac{97000}{8{,}314\cdot 298{,}15}} \quad\rightarrow\quad A = 2{,}43\cdot 10^{14}\ \mathrm{L^{3/2}\cdot mol^{-3/2}\cdot s^{-1}}$$ **2 mark**

Penalize with -1 mark if the units of the pre-exponential factor are incorrect.

The determination of reaction orders can also be performed using a redox titration. Unreacted potassium dichromate with alcohol can be determined with a standard solution of iron(II) sulfate. To carry out the study, a 100 mL volumetric flask (problem solution) was prepared where 60 mL of 1,40 M ethanol and 20,0 mL of 0,100 M potassium dichromate were added and the total volume was completed with a 3,0 M solution for H_2SO_4. Then, samples of 10,0 mL of the problem solution were taken every 30 s and titrated with an acidified standard solution of 0,0900 M Fe^{2+}, using 1,1,-phenanthroline as redox indicator. Before beginning each titration, a small amount of phosphoric acid was added to avoid color interference from the Fe^{3+} ions that form.

$$Cr_2O_7^{2-}(ac) + 6Fe^{2+}(ac) + 14H_3O^+(ac) \rightarrow 2Cr^{3+}(ac) + 6Fe^{3+}(ac) + 21H_2O(l)$$

6.6. Indicate in the figure the laboratory tools used in a titration.



4 mark in total (1 mark for identifying each tool in the figure).

6.7. Calculate the initial pH of the problem solution. $pK_{a2}(H_2SO_4) = 1{,}99$.

Note: If you cannot solve the previous item, consider pH = 0,20 to answer the next question.

$$c(H_2SO_4) = \frac{20}{100}\cdot 3{,}0 = 0{,}60\ \mathrm{mol\cdot L^{-1}}$$ **1 mark**

$$HSO_4^-(ac) \rightleftharpoons SO_4^{2-}(ac) + H^+(ac)$$

c_{eq} $\quad 0{,}6-x \quad\quad x \quad\quad 0{,}6+x$ **2 mark**

$$K_{a2} = 1{,}02\cdot 10^{-2} = \frac{[SO_4^{2-}][H^+]}{[HSO_4^-]} = \frac{(x)(0{,}6+x)}{0{,}6-x} \quad\rightarrow\quad x = 9{,}87\cdot 10^{-3}$$ **1 mark**

$[H^+] = 0{,}61\ \mathrm{mol\cdot L^{-1}}$ **1 mark** $\quad pH = -\log 0{,}61 = 0{,}21$ **1 mark**

6.8. Estimate the value of the potential at the equivalence point of the first titration (t = 0 s, see the table with the volume of solution consumed). $E^o(Fe^{3+}/Fe^{2+}) = +0{,}77$ V, $E^o(Cr_2O_7^{2-}/Cr^{3+}) = +1{,}33$ V.

Answer 1

Since the reaction is practically quantitative, at equilibrium we have

$$n(Fe^{3+})_{final} = n(Fe^{2+})_{reac} = 13{,}33\cdot 0{,}09 = 1{,}20\ \mathrm{mmol}$$ **1 mark**

$$c(Fe^{3+})_{final} = \frac{n(Fe^{3+})_{final}}{V(total)} = \frac{1{,}20}{13{,}33+10{,}0} = 0{,}0514\ \mathrm{mol\cdot L^{-1}}$$ **1 mark**

$n(Cr^{3+})_{final} = 2 \cdot n(Cr_2O_7^{2-})_{reac} = 2 \cdot \frac{1}{6} \cdot n(Fe^{2+})_{reac} = 2 \cdot \frac{1}{6} \cdot 1{,}20 = 0{,}400\ \text{mmol}$ **2 mark**

$c(Cr^{3+})_{final} = \frac{n(Cr^{3+})_{final}}{V(total)} = \frac{0{,}400}{13{,}33+10{,}0} = 0{,}0171\ \text{mol} \cdot L^{-1}$ **1 mark**

$Cr_2O_7^{2-}(ac) + 14H^+(ac) + 6e^- \rightarrow 2Cr^{3+}(ac) + 7H_2O(l)$

$E_{eq} = E_1° - \frac{0{,}0592}{6} \cdot \log\left(\frac{[Cr^{3+}]^2}{[H^+]^{14}[Cr_2O_7^{2-}]}\right)$ (1) **1 mark**

$Fe^{3+}(ac) + e^- \rightarrow Fe^{2+}(ac)$

$E_{eq} = E_2° - \frac{0{,}0592}{1} \cdot \log\left(\frac{[Fe^{2+}]}{[Fe^{3+}]}\right)$ (2) **1 mark**

Adding the equations (1) and (2), we get

$(6+1) \cdot E = 6 \cdot E_1° - 0{,}0592 \cdot \log\left(\frac{[Cr^{3+}]^2}{[H^+]^{14}[Cr_2O_7^{2-}]}\right) + 1 \cdot E_2° - 0{,}0592 \cdot \log\left(\frac{[Fe^{2+}]}{[Fe^{3+}]}\right)$ **1 mark**

But at equilibrium, it is true that

$Cr_2O_7^{2-}(ac) + 6Fe^{2+}(ac) + 14H^+(ac) \rightarrow 2Cr^{3+}(ac) + 6Fe^{3+}(ac) + 7H_2O(l)$

$[Fe^{2+}] = 6 \cdot [Cr_2O_7^{2-}]$ **1 mark**

The concentration of sulfuric acid and therefore hydronium ions, also changes

$$c_{dil}(H_2SO_4) = 0{,}60 \cdot \frac{10}{10 + 13{,}33} = 0{,}26\ \text{mol} \cdot L^{-1}$$

$$1{,}02 \cdot 10^{-2} = \frac{[SO_4^{2-}][H^+]}{[HSO_4^-]} = \frac{(x)(0{,}26 + x)}{0{,}26 - x} \rightarrow [H^+] = 0{,}27\ \text{mol} \cdot L^{-1}$$ **2 mark**

$(6+1) \cdot E_{eq} = 6 \cdot E_1° + 1 \cdot E_2° - 0{,}0592 \cdot \log\left(\frac{[Cr^{3+}]^2[Fe^{2+}]}{[H^+]^{14}[Cr_2O_7^{2-}][Fe^{3+}]}\right)$

$(6+1) \cdot E_{eq} = 6 \cdot E_1° + 1 \cdot E_2° - 0{,}0592 \cdot \log\left(\frac{[Cr^{3+}]^2 \cdot 6 \cdot [Cr_2O_7^{2-}]}{[H^+]^{14}[Cr_2O_7^{2-}][Fe^{3+}]}\right)$

$(6+1) \cdot E_{eq} = 6 \cdot 1{,}33 + 1 \cdot 0{,}77 - 0{,}0592 \cdot \log\left(\frac{0{,}0171^2 \cdot 6}{0{,}27^{14} \cdot 0{,}0514}\right)$

$E_{eq} = \frac{6 \cdot 1{,}33 + 1 \cdot 0{,}77 - 0{,}384}{6+1} = 1{,}20\ V$ **2 mark**

Answer 2

If the calculation is made by the formula for stoichiometric quantities directly, without taking into account the correction of the concentrations, only **4 mark** are awarded.

$$E_{eq} = \frac{n_1 \cdot E_1° + n_2 \cdot E_2°}{n_1 + n_2} = \frac{6 \cdot E_1° + 1 \cdot E_2°}{6 + 1} = \frac{6 \cdot 1{,}33 + 1 \cdot 0{,}77}{6 + 1} = 1{,}25\ V$$

Under these conditions, when the concentrations of alcohol and hydronium ions are much higher than the concentration of dichromate, it can be assumed that the reaction rate depends only on the concentration of the latter, since the other concentrations remain practically constant and then $v_r = k_r' \cdot [Cr_2O_7^{2-}]^y$, where $k_r' = k_r \cdot [RCH_2OH]^x [H_3O^+]$. Mathematically, it can be shown that depending on the value of the partial reaction order of the dichromate (y) the following equations stand

Order	$y = 1$	$y \neq 1$
Rate law	$\ln\left(\frac{[Cr_2O_7{}^{2-}]}{c_0(Cr_2O_7{}^{2-})}\right) = -k_r{}' \cdot t$	$[Cr_2O_7{}^{2-}]^{(1-y)} = -(1-y)\cdot k_r{}'\cdot t + c_0(Cr_2O_7{}^{2-})^{(1-y)}$

The volume of iron(II) sulfate needed to reach the equivalence point in each case was

Time (s)	0	30	60	90	120	150
Volume of Fe^{2+} (mL)	13,33	9,76	6,75	4,29	2,39	1,04

6.9. Calculate the concentration of dichromate at each time and complete the graph to show that the partial reaction order is y = ½ in this case.

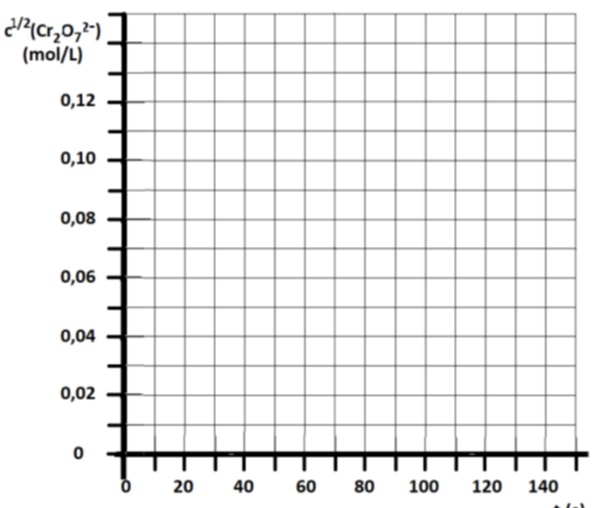
$$c(Cr_2O_7{}^{2-}) = \frac{n(Cr_2O_7{}^{2-})}{V(Cr_2O_7{}^{2-})} = \frac{\frac{1}{6}\cdot n(Fe^{2+})}{V(Cr_2O_7{}^{2-})} = \frac{1}{6}\cdot\frac{c(Fe^{2+})\cdot V(Fe^{2+})}{V(Cr_2O_7{}^{2-})} = \frac{1}{6}\cdot\frac{0{,}0900\cdot V(Fe^{2+})}{10{,}0}$$ **2 mark**

According to the data in the problem, the reaction order is y = ½. Therefore,

$$c^{1/2}(Cr_2O_7{}^{2-}) = -\frac{1}{2}\cdot k_r{}'\cdot t + c_0{}^{1/2}(Cr_2O_7{}^{2-})$$

t (s)	0	30	60	90	120	150
$c(Fe^{2+})$ (mL)	13,33	9,76	6,75	4,29	2,39	1,04
$c(Cr_2O_7{}^{2-})$ (mol/L)	0,02000	0,01464	0,01013	0,006435	0,003585	0,00156
$c^{1/2}(Cr_2O_7{}^{2-})$	0,1414	0,1210	0,1006	0,08022	0,05987	0,03950

3 mark



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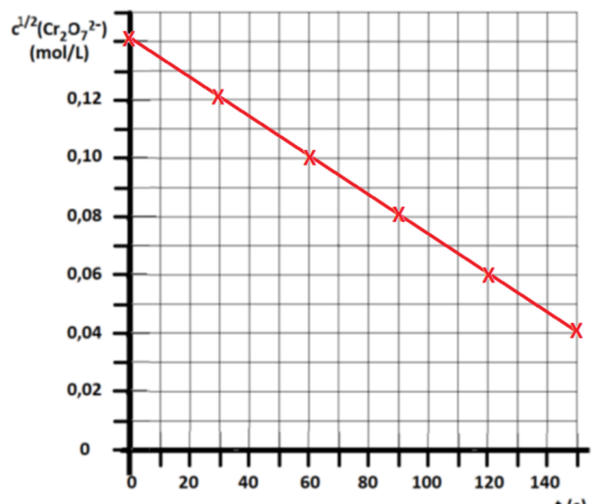

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5 mark

6.10. Using the graph you constructed or using a linear regression calculation, calculate the value of the rate constant k_r' and the partial reaction order x corresponding to the alcohol. Use the values of k_r and $[H_3O^+]$ you calculated in **6.2** and **6.7**, respectively.

Answer 1

Using a linear regression calculation, we obtain

$r = -1{,}0000 \quad \text{slope} = -6{,}793\cdot 10^{-4} \quad \text{intercept} = 0{,}1414$

$-6{,}793\cdot 10^{-4} = -\frac{1}{2}\cdot k_r{}' \quad \rightarrow \quad k_r{}' = 1{,}36\cdot 10^{-3}$ **3 mark**

$c_0(RCH_2OH) = \frac{60}{100} \cdot 1{,}40 = 0{,}840\ mol \cdot L^{-1}$ **1 mark**

$k_r' = k \cdot [RCH_2OH]^x \cdot [H_3O^+]$

$[RCH_2OH]^x = \frac{k_r'}{k \cdot [H_3O^+]} = \frac{1{,}36 \cdot 10^{-3}}{2{,}48 \cdot 10^{-3} \cdot 0{,}61} = 0{,}899$ **1 mark**

$0{,}840^x = 0{,}899 \quad \rightarrow \quad x \approx 1$ **1 mark**

Answer 2

It is possible to calculate the values of k_r' for each time and then the average value.

$k_r' = \frac{-2 \cdot [c^{1/2} - c_0{}^{1/2}]}{t}$ **1 mark**

t (s)	0	30	60	90	120	150
k_r'	0,00136	0,00136	0,00136	0,00136	0,00136	0,00136

2 mark

$c_0(RCH_2OH) = \frac{60}{100} \cdot 1{,}40 = 0{,}840\ mol \cdot L^{-1}$ **1 mark**

$k_r' = k \cdot [RCH_2OH]^x \cdot [H_3O^+]$

$[RCH_2OH]^x = \frac{k_r'}{k \cdot [H_3O^+]} = \frac{1{,}36 \cdot 10^{-3}}{2{,}48 \cdot 10^{-3} \cdot 0{,}61} = 0{,}899$ **1 mark**

$0{,}840^x = 0{,}899 \quad \rightarrow \quad x \approx 1$ **1 mark**

Calculating the slope from the graph,the range $0{,}0100 < k_r' < 0{,}0170$ is acceptable.

The determination of dichromate can also be carried out indirectly by iodometry, using starch as indicator. A fraction of the problem solution can be reacted with excess iodide and the triiodide ion formed is then titrated with a thiosulfate solution of known concentration.

6.11. Write balanced ionic equations for all reactions that occur during the determination of dichromate by iodometry.

$Cr_2O_7{}^{2-}(ac) + 6I^-(ac) + 14H^+(ac) \rightarrow 2Cr^{3+}(ac) + 3I_2(ac) + 7H_2O(l)$ **2 mark**

$I_2(ac) + I^-(ac) \rightarrow I_3{}^-(ac)$ **1 mark**

$2S_2O_3{}^{2-}(ac) + I_3{}^-(ac) \rightarrow S_4O_6{}^{2-}(ac) + 3I^-(ac)$ **2 mark**

Problem 7. Cryoscopy (12th grade)

7.1	7.2	7.3	7.4	7.5	Total	Total
4	7	4	12	13	40	10%
4	7	4	12	13	40	10%

Cryoscopy is one of the classical techniques for the characterization of compounds. By complete combustion, it was determined that a certain monocarboxylic acid **A** (of general formula R-COOH) has 40,0% carbon and 6,71% H. Three solutions were prepared by mixing 15,0 g of **A** in 250,0 g of different solvents. Next, the freezing temperatures of the pure solvents and those of the solutions was measured and the specific weight of each solution was determined with a hydrometer. The data was collected in the table below. It is known that the behavior of **A** in each solvent is different. While in 1,2-dibromoethane ($BrCH_2CH_2Br$) **A** behaves as a nonelectrolyte, in benzene **A** molecules associate to form an A_2 dimer that is stabilized by hydrogen bonds, and in water **A** dissociates into ions.

Solvent	T^o_c pure solvent (ºC)	T_c solution of **A** (ºC)	K_c (ºC·kg/mol)	Solvent density (g/mL)
benzene	5,53	2,49	5,12	0,878
1,2-dibromoethane	9,97	–2,53	12,5	2,18
water	0,00	–1,87	1,86	1,00

7.1. Relate the polarity of the solvent with the behavior of **A** in the different solutions.

In very polar solvents the acid dissociates. **2 mark**

In very nonpolar solvents the acid associates. **2 mark**

7.2. Using the percent composition offered, show that **A** is acetic acid.

Assuming 100 g of **A**

$n(C) = \frac{m(C)}{A_r(C)} = \frac{40,0}{12,0} = 3,33 \text{ mol}$ **1 mark**

$n(H) = \frac{m(H)}{A_r(H)} = \frac{6,71}{1,01} = 6,64 \text{ mol}$ **1 mark**

$n(O) = \frac{m(C)}{A_r(C)} = \frac{100-40,0-6,71}{12,0} = 3,33 \text{ mol}$ **1 mark**

$n(C) : n(H) : n(O) = 3,33 : 6,64 : 3,33 = 1 : 2 : 1 \quad \rightarrow \quad CH_2O$ **2 mark**

Since it is known to be a monocarboxylic acid, it has two oxygen atoms and the formula would be

$C_2H_4O_2 \equiv CH_3COOH$ acetic acid **2 mark**

7.3. Using the data from dissolving **A** in 1,2-dibromoethane, calculate the molar mass of the acid.

In 1,2-dibromoethane, acid **A** behaves as a nonelectrolyte. Therefore,

$\Delta T_{cry} = K_{cry} \cdot b = K_{eb} \cdot \frac{n(\mathbf{A})}{m(d)} \cdot 1000 = \frac{K_{cry} \cdot m(\mathbf{A}) \cdot 1000}{M(\mathbf{A}) \cdot m(d)}$ **1 mark**

$\Delta T_{cry} = 9,97 - (-2,53) = 12,5\ ℃$ **1 mark**

$12,5 = \frac{12,5 \cdot 15,0 \cdot 1000}{M(\mathbf{A}) \cdot 250} \quad \rightarrow \quad M(\mathbf{A}) = 60,0 \text{ g} \cdot \text{mol}^{-1}$ **2 mark**

This result confirms the identity of the acid.

7.4. Estimate the acid dissociation constant of **A** in water. Formulate the equilibrium that is established.

$CH_3COOH(ac) \rightleftharpoons CH_3COO^-(ac) + H^+(ac)$ **1 mark**

$b = \frac{m(\mathbf{A})\cdot 1000}{M(\mathbf{A})\cdot m(d)} = \frac{15{,}0\cdot 1000}{60{,}01\cdot 250{,}0} = 1{,}00\ mol\cdot kg^{-1}$ **1 mark**

$\Delta T_{cry} = 0 - (-1{,}87) = 1{,}87\ °C$ **1 mark**

$\Delta T_{cry} = i\cdot K_{cry}\cdot b$

$1{,}87 = 1{,}86\cdot i\cdot 1{,}00 \quad \rightarrow \quad i = 1{,}0054$ **1 mark**

$\alpha = \frac{i-1}{v-1} = \frac{1{,}0054-1}{2-1} = 0{,}0054$

$V(H_2O) = \rho(H_2O)\cdot m(H_2O) = 1{,}00\cdot 250 = 250\ mL$

$c_0(CH_3COOH) = \frac{n(CH_3COOH)}{V(H_2O)} = \frac{0{,}250}{0{,}250} = 1{,}00\ mol\cdot L^{-1}$ **2 mark**

$[H^+] = [CH_3COO^-] = \alpha\cdot c_0(CH_3COOH) = 0{,}0054\cdot 1{,}00 = 0{,}0054\ mol\cdot L^{-1}$ **2 mark**

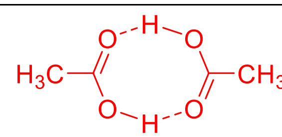
$[CH_3COOH] = (1-\alpha)\cdot c_0(CH_3COOH) = (1-0{,}0054)\cdot 1{,}00 = 0{,}9946\ mol\cdot L^{-1}$ **2 mark**

$K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]} = \frac{0{,}0054^2}{0{,}9946} = 2{,}9\cdot 10^{-5}$ **2 mark**

Slightly different values may be obtained depending on rounding.

7.5. Calculate the association constant of **A** in benzene. Formulate the equilibrium that is established and represent the structure of the $\mathbf{A}_2$ dimer.

$2CH_3COOH(solv) \rightleftharpoons (CH_3COOH)_2(solv)$ **1 mark**



2 marca

$\Delta T_{cry} = 5{,}53 - 2{,}49 = 3{,}04\ °C$ **1 mark**

$\Delta T_{cry} = i\cdot K_{cry}\cdot b$

$3{,}04 = 5{,}12\cdot i\cdot 1{,}00 \quad \rightarrow \quad i = 0{,}594$ **1 mark**

$V(benzene) = \rho(benzene)\cdot m(benzene) = 0{,}878\cdot 250 = 219{,}5\ mL$

$c_0(CH_3COOH) = \frac{n(CH_3COOH)}{V(H_2O)} = \frac{0{,}250}{0{,}2195} = 1{,}139\ mol\cdot L^{-1}$ **2 mark**

$$
\begin{array}{lll}
 & 2CH_3COOH(solv) \rightleftharpoons & (CH_3COOH)_2(solv) \\
c_{eq} & 1{,}139\cdot(1-\alpha) & \frac{\alpha}{2}\cdot 1{,}139
\end{array}
$$

$i = \frac{n_{tot}}{n_0} = \frac{1{,}139(1-\alpha/2)}{1{,}139} \quad \rightarrow \quad \alpha = 0{,}812$

$[CH_3COOH] = c_0(CH_3COOH)\cdot(1-\alpha) = 1{,}139\cdot(1-0{,}812) = 0{,}214\ mol\cdot L^{-1}$ **2 mark**

$[(CH_3COOH)_2] = c_0(CH_3COOH)\cdot \alpha/2 = 1{,}139\cdot 0{,}812/2 = 0{,}462\ mol\cdot L^{-1}$ **2 mark**

$K_a = \frac{[CH_3COOH]}{[(CH_3COOH)_2]^2} = \frac{0{,}462}{0{,}214^2} = 10{,}1$ **2 mark**

Quite different values can be obtained depending on rounding. Maximum punctuation is awarded if the procedure is correct and there are no calculation errors.

Problem 8. Miscellanea (10th grade)

8.1	…	Total	Total
4	idem	40	10%
4	idem	40	10%

Select the most appropriate statement in each case. You do not need to justify your answer.

8.1. The compound whose 1,00 g sample contains the largest number of atoms is:
A) CO B) C_2H_6 C) C_6H_6 D) Kr

8.2. Which species has the smallest bond angle?
A) NO_3^- B) NO_2^- C) FNO_2 D) NF_3

8.3. Which solid has the highest melting point?
A) S_8 B) MgF_2 C) SiO_2 D) P_4

8.4. What is the molar mass of a gas with density equal to 6,13 g/L at 25 °C and 101,325 kPa?
A) 10 g/mol B) 20 g/mol C) 150 g/mol D) 190 g/mol

8.5. The lattice energy of Na_2O has a value of –2600 kJ/mol, while the lattice energy of another oxide MgO has a value of –3900 kJ/mol. What factors contribute to MgO having a higher lattice energy?
I) Mg^{2+} has a charge larger than Na^+, II) Mg^{2+} is smaller than Na^+
A) only I B) only II C) I and II D) nor I nor II

8.6. What metal is resistant to rust in the open air because it forms a protective layer of oxide?
A) Al B) Ca C) Ag D)Sr

8.7. Which molecule has the shortest C-N bond distance?
A) HCNO B) HNCO C) H_3CNO D) H_3CNO_2

8.8. In the complex compound $Na[V(CO)_5]$, what is the oxidation number of vanadium?
A) 1– B) 3+ C) 5+ D) 6+

8.9. What is the coefficient for Zn when the equation is balanced to its minimum coefficients?
$Zn + H^+(ac) + NO_3^-(ac) \longrightarrow Zn^{2+}(ac) + N_2O(g) + H_2O(l)$
A) 2 B) 4 C) 6 D) 8

8.10. Increasing order of acid strength:
A) $H_3BO_3 < H_3PO_4 < HClO_3$ B) $HClO_3 < H_3BO_3 < H_3PO_4$
C) $H_3PO_4 < HClO_3 < H_3BO_3$ D) $H_3BO_3 < HClO_3 < H_3PO_4$

4 marks are awarded for each correct answer.

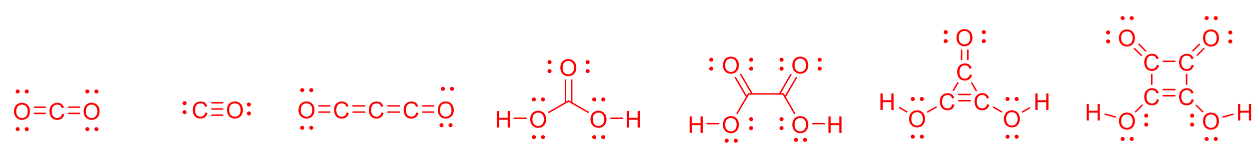
Problem 9. Carbon compounds (10th grade)

9.1	9.2	9.3	9.4	9.5	9.6	9.7	Total	Total
15	8	13	5	4	4	11	60	15%
15	8	13	5	4	4	11	60	15%

Carbon is an essential element for life. Perhaps its best-known chemical property is to form long chains in organic molecules, however, this element also has rich inorganic chemistry. Different oxides of carbon are known, including monoxide, dioxide, and suboxide (C_3O_2). In these molecules the central atom is carbon and they are completely linear. Carbonic acid is unstable and in the presence of water it decomposes rapidly to form dissolved carbon dioxide. In basic aqueous medium, carbon dioxide loses its protons forming hydrogencarbonate and carbonate anions. Deltic acid has the formula $C_3O(OH)_2$ and it is also called trianglic acid, in reference to its highly symmetrical structure with a three-atom cycle. Squaric acid has the formula $C_4O_2(OH)_2$ and it is also called quadratic acid, in reference to its highly symmetrical structure with a four-atom cycle. Both acids are relatively strong, therefore, they lose two protons easily, originating the corresponding dianions, which are aromatic. Another important carbon acid is oxalic $H_2C_2O_4$, which is used as primary standard in redox titrations because it is readily oxidized to carbon dioxide.

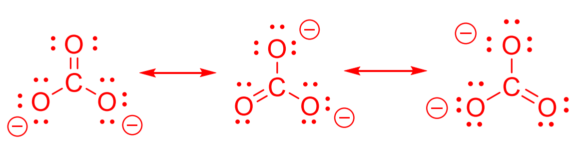
Carbon compounds exist for most chemical elements with diverse stoichiometry. Ionic carbides originate with metals and they can be of different types. Methanides (C^{4-}) such as Li_4C are derivatives of methane. Acetylene or ethyne can react with various metals, releasing dihydrogen and forming the corresponding acetylides (C_2^{2-}), such as CaC_2. Allenides or sesquicarbides (C_3^{4-}) such as Mg_2C_3 and interstitial compounds of graphite with alkali metals MC_{8n} are also known. All these carbides hydrolyze in water forming metal hydroxides and a hydrocarbon (compound formed exclusively by carbon and hydrogen). With the transition elements, carbon forms interstitial compounds, which are good electrical conductors; for example, cementite (6,68% by mass of C) is a major constituent of steel and cast iron. With nonmetals, carbon forms a wide variety of compounds.

9.1. Represent the structures for the three oxides of carbon mentioned, carbonic acid, oxalic acid, deltic acid, and squaric acid.



2 mark 2 mark 2 mark 2 mark 2 mark 2 mark 3 mark

9.2. Represent all the resonant structures of the carbonate CO_3^{2-}, oxalate $C_2O_4^{2-}$, deltate $C_3O_3^{2-}$, and squarate $C_4O_4^{2-}$ dianions.



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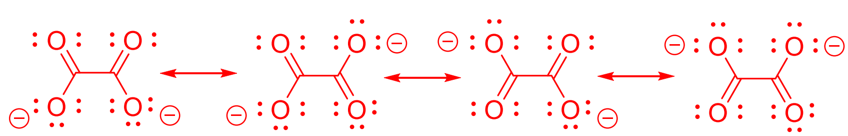

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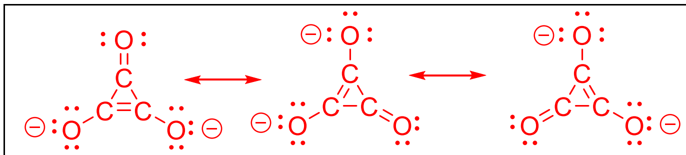

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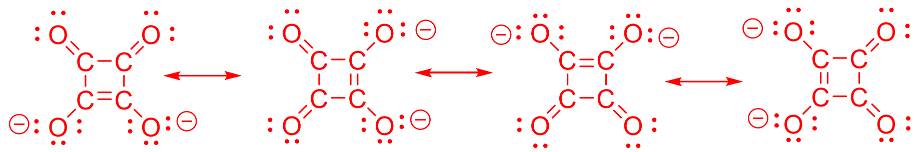


2 marca

9.3. Write balanced chemical equations for the following processes.

9.3.a) The decomposition of magnesium oxalate under intense heating forming only oxides.

$MgC_2O_4(s) \rightarrow MgO(s) + CO(g) + CO_2(g)$ **2 mark**

9.3.b) The reaction between potassium permanganate and oxalic acid, in the presence of sulfuric acid, if it is known to form manganese(II) sulfate and carbon dioxide.

$2KMnO_4(ac) + 5H_2C_2O_4(ac) + 3H_2SO_4(ac) \rightarrow K_2SO_4(ac) + 10CO_2(g) + 2MnSO_4(ac) + 8H_2O(l)$

3 mark

9.3.c) The preparation of calcium acetylide from calcium oxide.

$CaO(s) + 3C(s) \rightarrow CaC_2(ac) + CO(g)$ **2 mark**

Also acceptable: $CaO(s) + C_2H_2(g) \rightarrow CaC_2(ac) + H_2O(l)$ $Ca(s) + C_2H_2(g) \rightarrow CaC_2(ac) + H_2(g)$

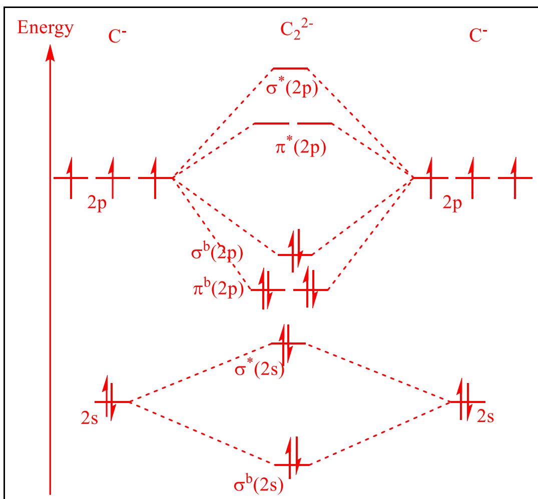
9.3.d) Hydrolysis of Li_4C, Na_2C_2 and Mg_2C_3.

$Li_4C(s) + 4H_2O(l) \rightarrow 4LiOH(ac) + CH_4(g)$ **2 mark**

$Na_2C_2(s) + 2H_2O(l) \rightarrow 2NaOH(s) + C_2H_2(g)$ **2 mark**

$Mg_2C_3(s) + 4H_2O(l) \rightarrow 2Mg(OH)_2(s) + C_3H_4(g)$ **2 mark**

9.4. Construct the molecular orbital diagram for the acetylide anion. Justify its formation by the calculation of the bond order. Predict its magnetic properties.



Reversing the order of the orbitals $\sigma^b(2p)$ and $\pi^b(2p)$ is acceptable. Using sp hybridization for both carbon C atoms is acceptable.

3 mark

Diamagnetic

1 mark

$$\text{Bond order} = \frac{8-2}{2} = 3$$

1 mark

9.5. Use valence bond theory and the hybridization model to justify the bond formation in ethyne molecule. (H–C≡C–H).

H $1s^1$
σ
C $2sp^1$ $2sp^1$ $2p^1$ $2p^1$
σ π π
C $2sp^1$ $2sp^1$ $2p^1$ $2p^1$
σ
H $1s^1$

4 mark

9.6. Find the empirical formula of cementite.

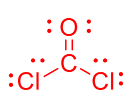
In 100 g of cementite,

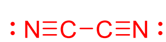
$n(C) = \frac{m(C)}{A_r(C)} = \frac{6{,}68}{12{,}0} = 0{,}557 \text{ mol}$ **1 mark**

$n(Fe) = \frac{m(Fe)}{A_r(Fe)} = \frac{100-6{,}68}{55{,}85} = 1{,}67 \text{ mol}$ **1 mark**

$n(C) : n(Fe) = 0{,}557 : 1{,}67 = 1 : 3$ **1 mark** → Fe_3C **1 mark**

9.7. Represent the structures of the phosgene compounds. $COCl_2$, cyanogen $(CN)_2$, cyanamide NH_2CN, dimethylcarbonate $CO(OCH_3)_2$ and methyl isocyanate CH_3NCO.


2 mark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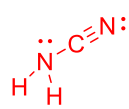
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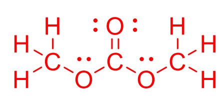
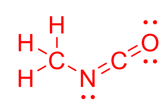

3 mark


2 mark

Other correct structures that correspond to the global formulas are acceptable.

Problem 10. Nickel extraction (10th grade)

10.1	10.2	10.3	10.4	Total	Total
8	6	23	3	40	10%

Cuba has important nickel deposits, mainly in the province of Holguín. This metal is widely used in the production of batteries. Nickel is found as part of laterites that are rich in iron and aluminum oxides. The composition of the mineral varies depending on the deposit in which it is found. The Caron process is used to isolate the component of interest from the rest of the matrix. Broadly speaking, the multi-stage process is summarized according to the following transformation ratio:

$$NiO \cdot Fe_2O_3 \cdot Al_2O_3(s) \rightarrow\rightarrow\rightarrow FeNi(s)$$

10.1. Determine the yield of the process if you start with 10,0 tons of laterite with empirical formula $NiO{\cdot}Fe_2O_3{\cdot}Al_2O_3$ and 1,20 ton of alloy with empirical formula FeNi are obtained. 1 ton = 10^3 kg = 10^6 g

The stoichiometry of the product implies that the limiting substance is nickel.

Answer 1

$M(NiO \cdot Fe_2O_3 \cdot Al_2O_3) = M(NiO) + M(Fe_2O_3) + M(Al_2O_3)$

$M(NiO \cdot Fe_2O_3 \cdot Al_2O_3) = 74{,}69 + 159{,}70 + 114{,}54 = 336{,}35\ g \cdot mol^{-1}$ **2 mark**

$n(FeNi)_{teor} = n(mineral) = \frac{m(mineral)}{M(mineral)} = \frac{10 \cdot 10^6}{336{,}35} = 29{,}73\ kmol$ **2 mark**

$m(FeNi)_{teor} = n(FeNi) \cdot M(FeNi) = 29{,}73 \cdot 10^3 \cdot 114{,}54 = 3{,}41\ ton$ **2 mark**

$Yield = \frac{n(FeNi)_{real}}{n(FeNi)_{teor}} \cdot 100 = \frac{1{,}20}{3{,}41} \cdot 100 = 35{,}2\%$ **2 mark**

Answer 2

$M(FeNi) = 55{,}85 + 58{,}69 = 114{,}54\ g \cdot mol^{-1}$ **1 mark**

$n(FeNi) = \frac{m(FeNi)}{M(FeNi)} = \frac{1{,}20 \cdot 1000}{114{,}54} = 10{,}48\ kmol$ **2 mark**

$M(NiO \cdot Fe_2O_3 \cdot Al_2O_3) = 74{,}69 + 159{,}70 + 114{,}54 = 336{,}35\ g \cdot mol^{-1}$ **1 mark**

$n(Ni) = n(mineral) = \frac{m(mineral)}{M(mineral)} = \frac{10 \cdot 10^6}{336{,}35} = 29{,}73\ kmol$ **2 mark**

$Yield = \frac{n(FeNi)}{n(mineral)} \cdot 100 = \frac{10{,}48}{29{,}73} \cdot 100 = 35{,}3\%$ **2 mark**

It has been identified that one of the stages that most affects the performance of the process is leaching, due to the solubility of nickel in the ammonia solution used. The equation that represents this loss of metal can be represented as:

$$Ni^{2+}(ac) + 4NH_3(ac) = [Ni(NH_3)_4]^{2+}(ac)$$

10.2. Determine the maximum mass of nickel that could be lost if 2,50 tons of ammonia are used to separate it from the ore.

$n(NH_3) = \frac{m(NH_3)}{M(NH_3)} = \frac{2{,}50 \cdot 10^6}{17{,}04} = 146{,}7\ kmol$ **2 mark**

$n(Ni) = \frac{1}{4} \cdot n(NH_3) = \frac{1}{4} \cdot 146{,}7 = 36{,}7\ kmol$ **2 mark**

$m(Ni) = A_r(Ni) \cdot n(Ni) = 58{,}69 \cdot 36{,}7 \cdot 10^3 = 2{,}15\ ton$ **2 mark**

More expensive alternatives would involve the use of an acid medium for separation.

10.3. Determine the mass of 30% by mass hydrochloric acid solution needed to dissolve 10,0 ton of laterite if its mass composition is NiO 1,0%, Fe_2O_3 (59,5%) and Al_2O_3 (11,5 %).

$NiO(s) + 2HCl(ac) \rightarrow NiCl_2(ac) + H_2O(l)$ **2 mark**

$m(NiO) = 0{,}1\ ton$ **1 mark** $n(NiO) = \frac{m(NiO)}{M(NiO)} = \frac{0{,}1 \cdot 10^6}{74{,}69} = 1{,}34\ kmol$ **1 mark**

$n(HCl)_{NiO} = 2 \cdot n(NiO) = 2 \cdot 1{,}34 = 2{,}68\ kmol$ **2 mark**

$Fe_2O_3(s) + 6HCl(ac) \rightarrow 2FeCl_3(ac) + 3H_2O(l)$ **2 mark**

$m(Fe_2O_3) = 5{,}95\ ton$ **1 mark** $n(Fe_2O_3) = \frac{m(Fe_2O_3)}{M(Fe_2O_3)} = \frac{5{,}95 \cdot 10^6}{159{,}70} = 37{,}26\ kmol$ **1 mark**

$n(HCl)_{Fe_2O_3} = 6 \cdot n(Fe_2O_3) = 6 \cdot 37{,}26 = 223{,}56\ kmol$ **2 mark**

$Al_2O_3(s) + 6HCl(ac) \rightarrow 2AlCl_3(ac) + 3H_2O(l)$ **2 mark**

$m(Al_2O_3) = 1{,}15\ ton$ **1 mark** $n(Al_2O_3) = \frac{m(Al_2O_3)}{M(Al_2O_3)} = \frac{1{,}15 \cdot 10^6}{101{,}96} = 11{,}28\ kmol$ **1 mark**

$n(HCl)_{Al_2O_3} = 6 \cdot n(Al_2O_3) = 6 \cdot 11{,}28 = 67{,}67\ kmol$ **2 mark**

$n(HCl)_{total} = n(HCl)_{NiO} + n(HCl)_{Fe_2O_3} + n(HCl)_{Al_2O_3}$

$n(HCl)_{total} = 2{,}68 + 223{,}56 + 67{,}67 = 293{,}91\ kmol$ **2 mark**

$m(HCl)_{total} = M(HCl) \cdot n(HCl)_{total} = 36{,}46 \cdot 293{,}91 = 10716\ kg = 10{,}72\ ton$ **1 mark**

$m(HCl)_{comercial} = \frac{100}{30} \cdot 10{,}72 = 35{,}72\ ton$ **2 mark**

10.4. How much would it cost to use hydrochloric acid in this process? The ton of industrial hydrochloric acid has a value of $38,000.

$cost = 35{,}72\ ton \cdot 38000\ \frac{\$}{ton} = \$\ 1{,}36 \cdot 10^6$ **3 mark**

M(X) in g/mol: NH_3 = 17,04; NiO = 74,69; Fe_2O_3 = 159,70; Al_2O_3 = 101,96; FeNi = 114,54; HCl = 36,46.

Problem 11. Thermochemistry of hydrocarbons (10th grade)

11.1	11.2	11.3	11.4	11.5	11.6	11.7	Total	Total
8	8	8	12	4	4	16	60	15%
8	8	8	12	4	4	16	60	15%

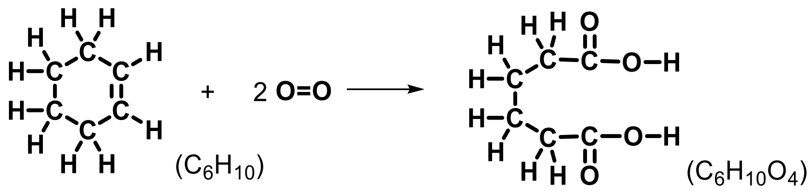
Cyclohexene (M = 82.14 g/mol) is a hydrocarbon that can react with potassium permanganate (M = 158.03 g/mol) according to the reaction

[1] $3C_6H_{10}(l) + 8KMnO_4(s) + 4H_2O(l) \rightarrow 3C_6H_{10}O_4(s) + 8MnO_2(s) + 8KOH(ac)$

The previous reaction can be divided for its thermodynamic study into three simpler reactions

[2] $4KMnO_4(s) + 2H_2O(l) \rightarrow 7O_2(g) + 4Mn(s) + 4KOH(ac) \qquad \Delta H_r° = 2123\ kJ/mol$

[3]



$\Delta H_r° = ¿?$

[4] $Mn(s) + O_2(g) \rightarrow MnO_2(s) \qquad \Delta H_r° = -520\ kJ/mol$

It is further known that $\Delta H_r° = \sum v \cdot \Delta H_f°$ $\qquad \Delta H_r° \approx \sum E(\text{broken bonds}) - \sum E(\text{formed bonds})$

11.1. Using the values of the binding energies, calculate the enthalpy change of the reaction **[3]**.

Bond	**O-H**	**C-C**	**C-H**	**C=C**	**C-O**	**O=O**	**C=O**
Energy (kJ/mol)	460	347	414	610	555	494	760

Note: If you could not solve the previous item, use the approximate value of $\Delta H_r°$**[3]** = –1100 kJ/mol to continue with the solution of the rest of the problem.

$\Delta H_r°[3] = 2 \cdot E(O=O) + E(C=C) + 2 \cdot E(C-H) - 2 \cdot E(C=O) - 2 \cdot E(C-O) - 2 \cdot E(O-H)$

6 mark (1 mark for each term)

$\Delta H_r°[3] = 2 \cdot 494 + 610 + 2 \cdot 414 - 2 \cdot 760 - 2 \cdot 555 - 2 \cdot 460 = -1124\ kJ \cdot mol^{-1}$ **2 mark**

It is acceptable to carry out the calculation with all the bonds present in reactants and products.

11.2. Using Hess' Law and enthalpy variations for reactions **[2]**, **[3]** and **[4]**, Calculate the enthalpy change for the reaction **[1]** between cyclohexene and potassium permanganate.

Note: If you could not solve the previous item, use the approximate value of $\Delta H_r°$**[1]** = –3300 kJ/mol to continue with the solution of the rest of the problem.

$4KMnO_4(s) + 2H_2O(l) \rightarrow 7O_2(g) + 4Mn(s) + 4KOH(ac) \quad /\cdot 2$ **2 mark**

$C_6H_{10}(s) + 2O_2(g) \rightarrow C_6H_{10}O_4(s) \quad /\cdot 3$ **2 mark**

$Mn(s) + O_2(g) \rightarrow MnO_2(s) \quad /\cdot 8$ **2 mark**

$3C_6H_{10}(s) + 8KMnO_4(s) + 4H_2O(l) \rightarrow 3C_6H_{10}O_4(s) + 8MnO_2(s) + 8KOH(ac)$

$\Delta H_r°[1] = \Delta H_r°[2] \cdot 2 + \Delta H_r°[3] \cdot 3 + \Delta H_r°[4] \cdot 8$

$\Delta H_r°[1] = 2123 \cdot 2 + (-1124) \cdot 3 + (-520) \cdot 8 = -3286\ kJ \cdot mol^{-1}$ **2 mark**

11.3. Calculate the heat evolved during the reaction of 13,85 g of cyclohexene with 60,3 g of potassium permanganate at atmospheric pressure.

$n(\text{hexene}) = \frac{m(\text{hexene})}{M(\text{hexene})} = \frac{13,85}{82,14} = 0,169\ \text{mol}$ **1 mark** $\frac{0,169}{3} = 0,0562\ \text{mol}$

$n(KMnO_4) = \frac{m(KMnO_4)}{M(KMnO_4)} = \frac{60,3}{158,03} = 0,382\ \text{mol}$ **1 mark** $\frac{0,382}{8} = 0,0477\ \text{mol}$ limiting substance

4 mark for identifying $KMnO_4$ as the limiting substance.

$Q_p = \xi \cdot \Delta H_r°[1] = 0,0477 \cdot (-3286) = -157\ \text{kJ}$ **2 mark**

11.4. Calculate the lattice energy of manganese(IV) oxide using the Born-Haber cycle, the energy values below, and the enthalpy of formation of the oxide (the enthalpy change of the reaction **[4]**).

$\Delta H_{sublim}°(Mn)$ = 234 kJ/mol; $I_1(Mn)$ = 717 kJ/mol; $I_2(Mn)$ = 1509 kJ/mol; $I_3(Mn)$ = 3248 kJ/mol; $I_4(Mn)$ = 4940 kJ/mol; $EA_1(O)$ = –141 kJ/mol; $EA_2(O)$ = 791 kJ/mol; $\Delta H_{dis}°(O_2)$ = 498 kJ/mol

$\Delta H_f°(MnO_2) = \Delta H_{sub}°(Mn) + I_1 + I_2 + I_3 + I_4 + 2 \cdot A_1 + 2 \cdot A_2 + \Delta H_{dis}°(O_2) + U(MnO_2)$

9 mark (1 mark for each term)

$U(MnO_2) = -520 - 234 - 717,4 - 1509,1 - 3248,4 - 4940 + 141 \cdot 2 - 791 \cdot 2 - 498$

$U(MnO_2) = -12967\ \text{kJ} \cdot \text{mol}^{-1}$ **3 mark**

11.5. The value obtained above is unrealistic because it was considered that Mn^{4+} cations and O^{2-} anions are present in the solid. Which chemical bond model best describes MnO_2? Justify your thinking.

The bond in manganese oxide has a large covalent character. **2 mark**

Manganese(IV) has a high charge/radius ratio and strongly polarizes the oxide ions, favoring the sharing of electrons, to the detriment of the formation of separated ions. **2 mark**

A 47,5 g sample of a certain alkane with the general formula C_nH_{2n+2} was completely combusted in a bomb calorimeter. As combustion products, only carbon dioxide and water were obtained and an energy of 2360 kJ was released, measured at constant pressure. The experiment was performed at 25,0 °C.

11.6. Estimate what the final temperature reached by the system was. It is known that the total heat capacity was equal to 1550 kJ/°C.

$\Delta T = \frac{Q_{abs}}{C} = \frac{-Q_{rel}}{C} = \frac{-(-2360)}{1550} = 1,52\ °C$ **2 mark**

$T_f = 25,0 + 1,5 = 26,5\ °C$ **2 mark**

11.7. Calculate the molecular formula of the alkane and its enthalpy of combustion in kJ/mol.

$\Delta H°_f\ (H_2O)$ = –286 kJ/mol, $\Delta H°_f\ (CO_2)$ = –394 kJ/mol, $\Delta H°_f$ (alkane) = –126 kJ/mol

$C_nH_{2n+2} + \frac{3n+1}{2} O_2 \rightarrow nCO_2 + (n+1)H_2O$ **2 mark**

$\Delta H_r° = n \cdot \Delta H_f°(CO_2) + (n+1) \cdot \Delta H_f°(H_2O) - \Delta H_f°(X)$

$\Delta H_r° = n \cdot (-394) + (n+1) \cdot (-286) - (-126)$

$\Delta H_r° = -680 \cdot n - 160$ **3 mark**

$\xi = \frac{n(X)}{v(X)} = \frac{m(X)}{M(X) \cdot v(X)} = \frac{47,5}{(n \cdot 12,0 + 2 \cdot n + 2) \cdot 1} = \frac{47,5}{14 \cdot n + 2}$ **2 mark**

$Q = \xi \cdot \Delta H_r{}^\circ$

$-2360 = \left(\frac{47,5}{14 \cdot n + 2}\right) \cdot (-680 \cdot n - 160)$ **2 mark**

$14 \cdot n + 2 = 13,7 \cdot n + 3,22$

$n = 3,94 \approx 4 \quad \rightarrow \quad C_4H_{10}$ **3 mark**

$\Delta H_r{}^\circ = -680 \cdot n - 160 = -680 \cdot 4 - 160 = -2880\ kJ \cdot mol^{-1}$ **4 mark**

Problem 12. Chemical equilibrium (only 11th and 12th grades)

12.1	12.2	12.3	12.4	12.5	Total	Total
9	9	8	7	7	40	10%
9	9	8	7	7	40	10%

12.1. An 18,92% by mass aqueous solution of a strong and soluble metal chloride was prepared. The change in the melting point of the solution from that of pure water was about 13,68 °C. Identify the metal chloride if it is known that K_{cryos} (H_2O) = 1,86 °C·kg·mol^{-1}.

$\Delta T_{eb} = i \cdot b \cdot K_{cryos} = i \cdot \frac{m(X)\cdot 1000 \cdot K_{crios}}{M(X)\cdot m(H_2O)} = i \cdot \frac{\omega_{\%}\cdot 1000 \cdot K_{cryos}}{M(X)\cdot[100-\omega_{\%}]}$ **2 mark**

Let the chloride have the formula MCl_n,

$i = n + 1$ y $M(MCl_n) = A_r(M) + n \cdot A_r(Cl)$

$$A_r(M) + n \cdot A_r(Cl) = (n+1) \cdot \frac{\omega_{\%} \cdot 1000 \cdot K_{cryos}}{\Delta T_{eb} \cdot [1-\omega_{\%}]}$$

$$A_r(M) = -n \cdot 35{,}45 + (n+1) \cdot \frac{18{,}92 \cdot 1000 \cdot 1{,}86}{13{,}68 \cdot [100 - 18{,}92]}$$

$$A_r(M) = -n \cdot 35{,}45 + 31{,}73 \cdot (n+1)$$

$A_r(M) = 31{,}73 - 3{,}72 \cdot n$ **5 mark**

Si $n = 1$ $A_r(M) = 28{,}0\ g \cdot mol^{-1}$ Si? SiCl?

Si $n = 2$ $A_r(M) = 24{,}3\ g \cdot mol^{-1}$ Mg $MgCl_2$ **2 mark**

12.2. Three gases A, B, and C are in equilibrium according to $A(g) \rightleftharpoons B(g) + C(g)$. A certain amount of A is introduced into an empty container. If the K_p at 298 K is 4,00 and the total pressure of the gases in equilibrium is 2,00 bar, calculate the ratio between the amounts of substance of B and of A in balance.

$K_p = K_X \cdot \left(\frac{p_{total}}{p^\circ}\right)^{\Delta v}$ $K_X = K_p \cdot \left(\frac{p^\circ}{p_{total}}\right)^{\Delta v} = 4{,}00 \cdot \left(\frac{1{,}00}{2{,}00}\right)^1 = 2{,}00$ **2 mark**

$A(g) \rightleftharpoons B(g) + C(g)$

In the equilibrium: $n_0 \cdot (1-x)$ $x \cdot n_0$ $x \cdot n_0$ $\rightarrow$ $n_{total} = n_0(1+x)$

$X_B = X_C$ **1 mark** $X_B = X_C = \frac{x \cdot n_0}{n_0 \cdot (1+x)} = \frac{x}{1+x}$ **2 mark** $X_A = \frac{n_0(1-x)}{n_0(1+x)} = \frac{1-x}{1+x}$ **2 mark**

$K_X = \frac{X_B \cdot X_C}{X_A} = \frac{\frac{x}{1+x}\cdot\frac{x}{1+x}}{\frac{1-x}{1+x}} = \frac{x^2}{1-x^2} = 2$ $\rightarrow$ $2 - 2x^2 = x^2$ $\rightarrow x = \sqrt{\frac{2}{3}} = 0{,}816$ **1 mark**

$\frac{n_B}{n_A} = \frac{x \cdot n_0}{n_0 \cdot (1-x)} = \frac{x}{1-x} = \frac{0{,}816}{1-0{,}816} = 4{,}45$ **1 mark**

12.3. Calculate the pH of the solution resulting from mixing 25 mL of NH_3 0,01 mol·L^{-1} with 15 mL of HCl 0,01 mol·L^{-1}. K_b (NH_3) = 1,8·10^{-5}.

$V_f = 25 + 15 = 40\ mL$

$NH_3(ac) + HCl(ac) \rightarrow NH_4Cl(ac)$

$n_0(NH_3) = c(NH_3) \cdot V(NH_3) = 0{,}01 \cdot 25 = 0{,}25\ mmol$ **1 mark**

$n_0(HCl) = c(HCl) \cdot V(HCl) = 0{,}01 \cdot 15 = 0{,}15\ mmol$ **1 mark** limiting substance **1 mark**

$n(NH_4Cl) = n_0(HCl) = 0{,}15\ mmol$ **1 mark**

$n_{exc}(NH_3) = n_0(NH_3) - n_0(HCl) = 0{,}25 - 0{,}15 = 0{,}10\ mmol$ **1 mark**

$$K_b = \frac{[NH_4{}^+] \cdot [OH^-]}{[NH_3]} \quad \rightarrow \quad [OH^-] = \frac{[NH_3] \cdot K_b}{[NH_4{}^+]} = \frac{\frac{0{,}10}{40} \cdot 1{,}8 \cdot 10^{-5}}{\frac{0{,}15}{40}} = 1{,}2 \cdot 10^{-5}\ mol \cdot L^{-1}$$

$pH = 14 - \log[OH^-] = 14 - \log(1{,}2 \cdot 10^{-5}) = 9{,}1$ **3 mark**

12.4. If the solubility of silver oxalate at 25 °C is $1{,}11 \cdot 10^{-4}\ mol \cdot L^{-1}$, calculate the solubility product of the salt. Neglect any hydrolysis of the oxalate ion.

$Ag_2C_2O_4(s) \rightleftharpoons 2Ag^+(ac) + C_2O_4{}^{2-}(ac)$

$K_{ps} = [Ag^+]^2 \cdot [C_2O_4{}^{2-}] = (2S)^2 \cdot S = 4 \cdot S^3$ **5 mark**

$K_{ps} = 4 \cdot (1{,}11 \cdot 10^{-4})^3 = 5{,}47 \cdot 10^{-12}$ **2 mark**

12.5. Calculate the solubility product constant of Hg_2Cl_2 in water at 25°C if the reduction redox potentials are $E°([Hg_2]^{2+}/Hg(l)) = 0{,}789\ V$ and $E°(Hg_2Cl_2(s)/Hg(l), Cl^-) = 0{,}268\ V$.

$Hg_2Cl_2(s) \rightleftharpoons Hg_2{}^{2+}(ac) + 2Cl^-(ac)$

$\Delta E° = E°(Hg_2Cl_2/Hg, Cl^-) - E°(Hg_2{}^{2+}/Hg) = 0{,}268 - 0{,}789 = -0{,}521\ V$ **5 mark**

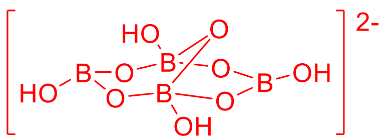
$K_{ps} = \exp\left(\frac{z \cdot F \cdot \Delta E°}{R \cdot T}\right) = \exp\left[\frac{2 \cdot (-0{,}521)}{0{,}0592}\right] = 2{,}44 \cdot 10^{-18}$ **2 mark**

Problem 13. Boric acid (11th grade)

13.1	13.2	13.3	13.4	13.5	13.6	13.7	13.8	13.9	13.10	13.11	13.12	13.13	Total	Total
3	3	4	8	7	6	6	3	5	2	9	11	13	80	20%
3	3	4	8	7	6	6	3	5	2	9	11	13	80	20%

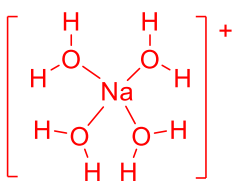
One of the laboratory practices of Experimental Chemistry at the University of Havana consists in preparing boric acid (H_3BO_3, M = 61,83 g/mol) from the reaction between the mineral borax or sodium tetraborate decahydrate ($Na_2B_4O_7 \cdot 10H_2O$, M = 381,37 g/mol, also formulated as $Na_2B_4O_5(OH)_4.8H_2O$) and hydrochloric acid. In this question we will use the results obtained by a Chemistry contestant in this practice in 2011.

13.1. Represent the structure of the anion $[B_4O_5(OH)_4]^{2-}$ present in borax. This ion is known to be symmetric and is formed by a bridged bicycle in which oxygen atoms alternate with boron atoms and one oxygen atom acts as a bridge between two boron atoms. Also, all the boron atoms are attached to a hydroxide group.



3 mark

13.2. Draw the structure of the cations present in borax, knowing that the remaining water molecules are covalently bound to sodium(I) ions.



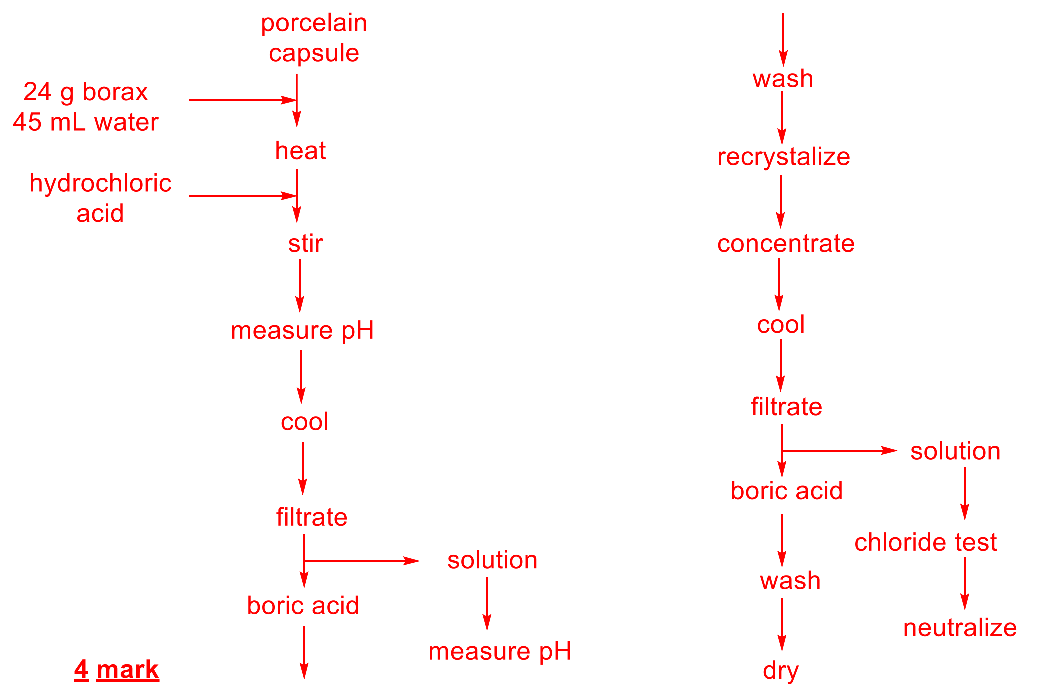
3 mark

The laboratory procedure used by the student was as follows.

1. 24,0 g of borax and 45,0 mL of water are added to a porcelain dish and gently heated under constant stirring until the solid is completely dissolved.
2. Add the hydrochloric acid in small portions to the hot borax solution, stirring constantly.
3. Once all the acid has been added, the acidity of the solution is tested, for which 2 or 3 drops of it are taken and methyl orange is added (it should take on a pink color). If the result is not positive, add hydrochloric acid in 1 mL portions until the result is positive.
4. Cool to 10 °C under stirring until as much boric acid as possible precipitates.
5. Filter the solid under reduced pressure.
6. The acidity of the filtrate is checked in a similar way to that described above. If the solution is not frankly acid, a sufficient amount of hydrochloric acid is added to achieve this, and it is filtered again. The boric acid obtained in this way is mixed with the initial found in the filter.
7. The total solid obtained is washed with 5-10 mL of cold water, well impregnating the crystals with the washing water before creating vacuum.
8. To purify the boric acid obtained by recrystallization, the solid is dissolved in the least amount of hot water, close to boiling temperature, and it is filtered by gravity while the solution is still hot.
9. The filtrate is poured into a 250 mL crystallizer, covered, and allowed to cool slowly.
10. Once crystallization is achieved, it is vacuum filtered to collect the crystals.
11. The filtrate from this last operation can be concentrated up to a third of its volume to obtain another portion of the product, which is filtered using the same paper that contains the previous solid portion.
12. The solid is washed with 5 ml of cold water repeatedly until the chlorides are removed. This is determined by testing with silver nitrate.
13. The pure crystals are dried on absorbent paper and weighed.

14. Finally, the filtrates or waste solutions are neutralized with sodium carbonate and disposed in the aqueous solution waste container.

13.3. Build a flowchart where the main operations performed by the student during the laboratory procedure are depicted and simplified.



4 mark

13.4. Write the balanced equation for the reaction between borax and hydrochloric acid (2 equivalents of the latter react). Calculate the yield of the process if it is known that the student obtained a final mass of 10,2 g of boric acid.

$Na_2B_4O_7 \cdot 10H_2O(s) + 2HCl(ac) \rightarrow 2NaCl(ac) + 4H_3BO_3(ac) + 5H_2O(l)$ **2 mark**

$n(H_3BO_3) = 4 \cdot n(borax)$ **2 mark**

$m(H_3BO_3)_{theorecal} = 4 \cdot \frac{m(borax)}{M(borax)} \cdot M(H_3BO_3) = \frac{4 \cdot 24{,}0}{381{,}37} \cdot 61{,}83 = 15{,}6\ g$ **2 mark**

$Yield = \frac{10{,}2}{15{,}6} \cdot 100 = 65\%$ **2 mark**

13.5. Calculate the minimum volume of commercial hydrochloric acid (37% by mass, 1,19 g/mL) that you used.

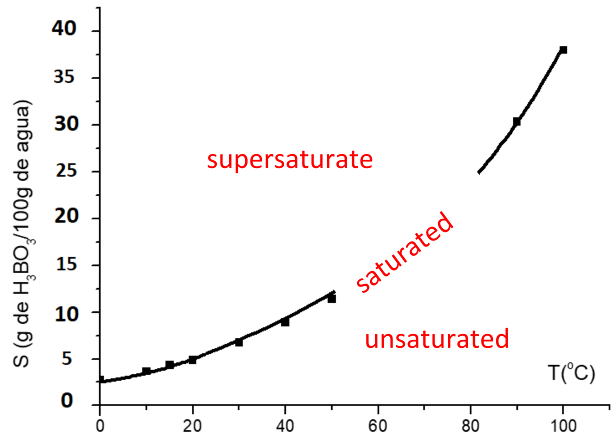
$n(HCl) = 2 \cdot n(borax) = 2 \cdot 0{,}0629 = 0{,}126\ mol$ **2 mark**

$M(HCl) = A_r(H) + A_r(H) = 1{,}01 + 35{,}45 = 36{,}46\ g \cdot mol^{-1}$ **1 mark**

$c(HCl) = \frac{w_{\%} \cdot d \cdot 10}{M(HCl)} = \frac{37 \cdot 1{,}19 \cdot 10}{36{,}46} = 12{,}1\ mol \cdot L^{-1}$ **2 mark**

$V(HCl) = \frac{n(HCl)}{c(HCl)} = \frac{0{,}126}{12{,}1} = 0{,}0104\ L \approx 10\ mL$ **2 mark**

13.6. This plot was drawn by the contestant. Indicate the zones that correspond to solutions: unsaturated, saturated and supersaturated.



Supersaturated: **2 mark**

Unsaturated: **2 mark**

Unsaturated: **2 mark**

13.7. Using the plot, calculate the total mass of dissolved boric acid that was lost during purification by recrystallization, if the mass of the solution was 45,0 g and its temperature 10 °C.

$S(10\ °C) \approx 3{,}0$ g per 100 g of water **3 mark**

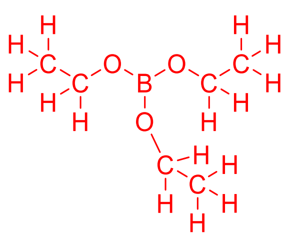
$\frac{S}{S+100} = \frac{m(H_3BO_3)}{45{,}0} = \frac{3{,}0}{3{,}0+100}$ **1 mark**

$m(H_3BO_3)_{loss} = 1{,}3$ g **2 mark**

The acceptable range of values are $2{,}0 \le S(10\ °C) \le 5$ and $0{,}9\ g \le m(H_3BO_3)_{loss} \le 2{,}1\ g$

To check the identity of the acid, X mixed a portion of the solid product, eight drops of ethanol, and four drops of concentrated sulfuric acid in a porcelain dish. Next, he stirred with a rod and then ignited the solution with a lighter. A green flame appeared, characteristic of the presence of boron compounds.

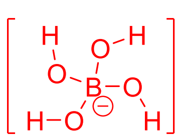
13.8. Represent the structure of the ethyl borate $B(OCH_2CH_3)_3$ formed.



3 mark

The semi-developed formula is accepted.

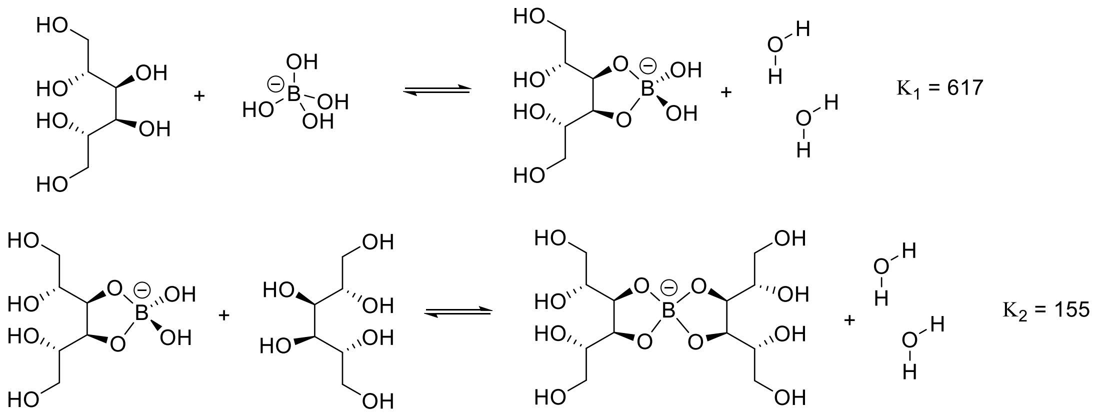
13.9. In aqueous solution boric acid behaves as monoprotic due to the formation of a tetracoordinated complex with a water molecule. Represent the dissociation equilibrium of boric acid and the structure of the tetrahydroxydoborate ion. $[B(OH)_4]^-$.



3 mark

$H_3BO_3(ac) + H_2O(l) \rightarrow [B(OH)_4]^-(ac) + H^+(ac)$ **2 mark**

Boric acid is very weak and its direct titration with sodium hydroxide is not possible. For this reason, it is necessary to add mannitol, a polyhydric alcohol capable of forming stable tetracoordinate complexes with boric acid, which causes its acidity to increase and the pH change near the equivalence point (and consequently, the jump in the pH curve). rating) is larger. Equilibria are reported in the literature.



13.10. Why do you think the second constant is four times smaller than the first?

The ligands are very bulky and cause high steric hindrance. **2 mark**

In order to determine the purity of the obtained boric acid, X proceeded to carry out an acid-base titration. He took 3,0905 g of the product and dissolved it in water to make a 500 mL solution in a volumetric flask. With a graduated pipette, 10,0 mL of the solution was transferred to a container. 1,0 g of mannitol, 30 mL of distilled water, and a few drops of phenolphthalein were added. It was then titrated with a 0,0963 mol/L sodium hydroxide solution, consuming an average volume of 10,5 mL.

13.11. Estimate the percent purity of the boric acid.

$H_3BO_3(ac) + NaOH(ac) \rightarrow Na[B(OH)_4](ac)$

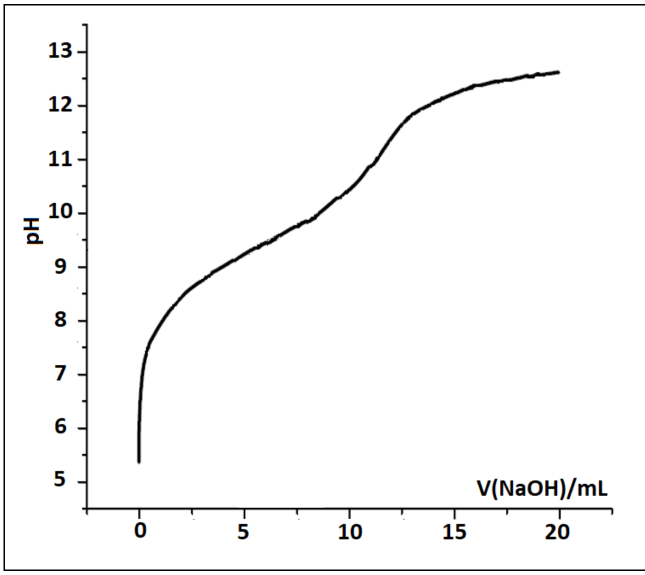
$n(NaOH) = c(NaOH) \cdot V(NaOH) = 0{,}0963 \cdot 10{,}5 = 1{,}011\ mmol$ **2 mark**

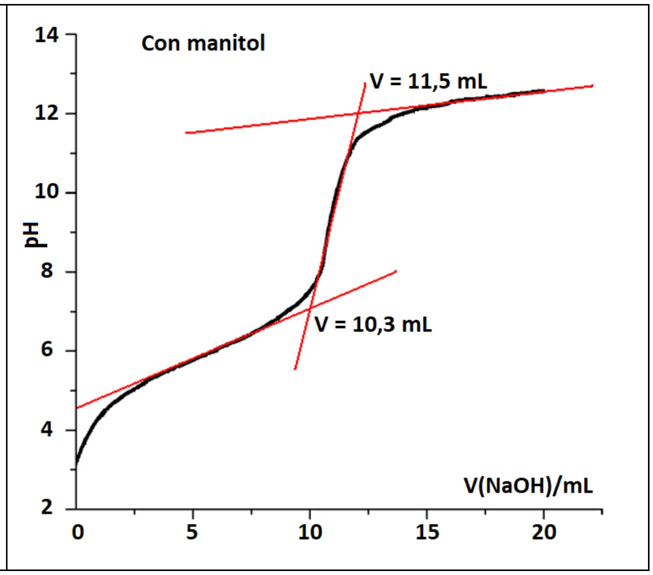
$n(H_3BO_3) = NaOH = 1{,}011\ mmol$ **2 mark**

$m(H_3BO_3) = n(H_3BO_3) \cdot M(H_3BO_3) \cdot \frac{500}{10} = 1{,}011 \cdot 10^{-3} \cdot 61{,}83 \cdot 50 = 3{,}1255\ g$ **2 mark**

$Purity = \frac{3{,}1255}{3{,}09} \cdot 100 = 101\%$ **1 mark** $\rightarrow$ 99% of purity **2 mark**

Immediately, our hero came up with a way to determine the formation constant of the complex between boric acid and mannitol. To do this, he repeated the previous assessment twice, with and without mannitol. This time, instead of using phenolphthalein, he used a pH-meter to build a titration curve (pH vs V) and determine the end point using the graphical method, with the help of the tangent method (the lines represented on the graph). The results obtained are shown in the figure.





13.12. Find the volume of equivalence using the data from the titration in the presence of mannitol and estimate the percent purity of the boric acid.

$V(NaOH)_{equiv} = \frac{11,5+10,3}{2} = 10,9\ mL$ **2 mark**

$n(NaOH) = c(NaOH) \cdot V(NaOH) = 0,0963 \cdot 10,9 = 1,05\ mmol$ **2 mark**

$n(H_3BO_3) = NaOH = 1,05\ mmol$ **2 mark**

$m(H_3BO_3) = n(H_3BO_3) \cdot M(H_3BO_3) \cdot \frac{500}{10} = 1,05 \cdot 10^{-3} \cdot 61,83 \cdot 50 = 3,2451\ g$ **2 mark**

$Purity = \frac{3,2451}{3,09} \cdot 100 = 105\%$ **1 mark** $\rightarrow$ 95% purity **2 mark**

13.13. Use the titration plots obtained by X to answer the following questions.

At the semiequivalence point pH = pK_a, in this case V = 5,45 mL. **3 mark**

13.13.a) Find the acid dissociation constant of boric acid in water.

$pK_a(H_3BO_3) = 9,31$ **3 mark** $\rightarrow$ $K_a = 10^{-9,31} = 4,90 \cdot 10^{-10}$ **1 mark**

A value in the interval is accepted $9,0 < pK_a < 9,6$.

(This value is quite close to the value reported in the literature for $pK_a = 9,24$.)

13.13.b) Find the apparent acid dissociation constant of boric acid in the presence of mannitol.

$pK_a'(H_3BO_3) = 5,85$ **3 mark** $\rightarrow$ $K_a' = 10^{-5,85} = 1,41 \cdot 10^{-6}$ **1 mark**

A value in the interval is accepted $5,55 < pK_a < 6,15$.

13.13.c) Estimate the formation constant of mannitol borate from the above constants.

$\beta = \frac{K_a'}{K_a} = \frac{1,41 \cdot 10^{-6}}{4,90 \cdot 10^{-10}} = 2,88 \cdot 10^3$ **2 mark**

This value is variable, according to what was obtained by the students in the previous paragraphs. The examiner must check that the calculation was correct.

Problem 14. Thermodynamics of carbon (11th grade)

14.1	14.2	14.3	14.4	14.5	14.6	14.7	14.8	14.9	14.10	14.11	14.12	14.13	14.14	14.15	Total	Total
8	2	5	3	2	6	6	7	2	7	6	10	3	6	7	80	20%
8	2	5	3	2	6	6	7	2	7	6	10	3	6	7	80	20%

Carbon dioxide is partially soluble in water because it reacts to form a small amount of carbonic acid, which dissociates into the corresponding anions.

[1] $CO_2(ac) + H_2O(l) \rightleftharpoons H_2CO_3(ac) \quad K = 1{,}70 \cdot 10^{-3}$

[2] $H_2CO_3(ac) \rightleftharpoons H^+(ac) + HCO_3^-(ac) \quad pK_{a1} = 3{,}60$

[3] $HCO_3^-(ac) \rightleftharpoons H^+(ac) + CO_3^{2-}(ac) \quad pK_{a2} = 10{,}25$

Species	$\Delta H_f°$ ($kJ \cdot mol^{-1}$)	S° ($J \cdot K^{-1} \cdot mol^{-1}$)
$CO_2(g)$	–393,5	213,7
$CO_2(ac)$	–412,9	121,3
C(s)	0	5,7
CO(g)	–110,5	197,6
CaO(s)	–635,5	39,7
$CaCO_3(s)$	–1207,0	88,7

Note: In this question, students can use either the values calculated by them, or the approximate values and data provided.

14.1. Show that the constant for the dissolution of carbon dioxide in water at 25°C is **approximately** equal to

[4] $CO_2(g) \rightleftharpoons CO_2(ac) \quad K_H = 3{,}50 \cdot 10^{-2}$

$\Delta H_r° = \Delta H_f°(CO_2, ac) - \Delta H_f°(CO_2, g) = -412{,}9 - (-393{,}5) = -19{,}4\ kJ \cdot mol^{-1}$ **2 mark**

$\Delta S_r° = S°(CO_2, ac) - S°(CO_2, g) = 121{,}3 - 213{,}7 = -92{,}4\ J \cdot K^{-1} \cdot mol^{-1}$ **2 mark**

$\Delta G_r° = \Delta H_r° - T \cdot \Delta S_r° = -19{,}4 \cdot 1000 - 298{,}15 \cdot (-92{,}4) = 8149{,}06\ J \cdot mol^{-1}$ **2 mark**

$K_H = \exp\left(\frac{\Delta G_r}{-RT}\right) = \exp\left(\frac{8149{,}06}{-8{,}314 \cdot 298{,}15}\right) = 3{,}74 \cdot 10^{-2}$ **2 mark**

14.2. If the rate constant for the forward reaction at equilibrium **[1]** is 0,039 s^{-1}, find the value of the rate constant for the reverse reaction.

$k_i = \frac{k_d}{K} = \frac{0{,}039}{1{,}70 \cdot 10^{-3}} = 22{,}9\ s^{-1}$ **2 mark**

14.3. The coefficient α (atm^{-1}) is defined as the volume of gas if it were measured at TPN (0 °C and 101,325 kPa) that is dissolved in the unit volume of solvent at a gas partial pressure equal to 1.00 atm. Using the given value of K_H, calculate the value of α at 25°C. V_m(TPN) = 22,4 L/mol for an ideal gas.

$[CO_2] = K_H \cdot p_r(CO_2) = 3{,}50 \cdot 10^{-2} \cdot \frac{101{,}325}{100} = 3{,}55 \cdot 10^{-2}\ mol/L$ **2 mark**

It is accepted to use 1,00 atm as standard pressure and use the value of K_H calculated in **14.1**.

In 1 L of aqueous solution there is approximately 1 L of water. Therefore,

$n(CO_2) = 3{,}55 \cdot 10^{-2}\ mol$ **1 mark**

$V(CO_2) = n(CO_2) \cdot V_m = 3{,}55 \cdot 10^{-2} \cdot 22{,}4 = 0{,}795\ L$ **1 mark**

$\alpha(CO_2) = 0{,}795\ atm^{-1}$ **1 mark**

Since the concentration of carbonic acid is small, **for all practical purposes and calculations** the first equilibrium of acid dissociation is considered only from dissolved carbon dioxide, according to

[5] $CO_2(ac) + H_2O(l) \rightleftharpoons H^+(ac) + HCO_3^-(ac) \quad pK_{a1}' = ???$

14.4. Show that the value of the acid dissociation constant of carbon dioxide is $pK_{a1}' = 6.37$.

Equilibrium [5] is obtained by adding equilibrium [1] and [2]. Therefore, **1 mark**

$pK_{a1}' = -\log K + pK_{a1} = -\log(1{,}70 \cdot 10^{-3}) + 3{,}70 = 2{,}76 + 3{,}60 = 6{,}36$ **2 mark**

Carbonated beverages consist of aqueous solutions of carbon dioxide at high pressure, in the presence of different acidulants and flavorings. Usually, the pH of these preparations is close to 3,0.

14.5. Why does opening a soda can of a carbonated drink cause fizz?

Because the partial pressure of the gas decreases. **2 mark**

14.6. Calculate the fractions of carbon dioxide, carbonic acid, and their two anions at pH equal to 3,0.

$[H^+]^2 + K_{a1}' \cdot [H^+] + K_{a1}' \cdot K_{a2} = (10^{-3})^2 + 10^{-3} \cdot 10^{-6{,}37} + 10^{-6{,}37} \cdot 10^{-10{,}25} = 1 \cdot 10^{-6}$ **1 mark**

$\alpha_0 = \frac{[CO_2]}{c(CO_2)_{total}} = \frac{[H^+]^2}{1{,}00 \cdot 10^{-6}} = \frac{1{,}00 \cdot 10^{-6}}{1{,}00 \cdot 10^{-6}} = 1{,}00$ **1 mark**

$\alpha_1 = \frac{[HCO_3^-]}{c(CO_2)_{total}} = \frac{K_{a1}' \cdot [H^+]}{1{,}00 \cdot 10^{-6}} = \frac{4{,}27 \cdot 10^{-10}}{1{,}00 \cdot 10^{-6}} = 4{,}27 \cdot 10^{-4}$ **1 mark**

$\alpha_2 = \frac{[CO_3^{2-}]}{c(CO_2)_{total}} = \frac{K_{a1}' \cdot K_{a2}}{1{,}00 \cdot 10^{-6}} = \frac{2{,}40 \cdot 10^{-17}}{1{,}00 \cdot 10^{-6}} = 2{,}40 \cdot 10^{-11}$ **1 mark**

$\frac{[H_2CO_3]}{c(CO_2)_{total}} = \frac{K \cdot [CO_2]}{1{,}00} = \frac{1{,}70 \cdot 10^{-3} \cdot 1{,}00 \cdot 10^{-6}}{1{,}00 \cdot 10^{-6}} = 1{,}70 \cdot 10^{-3}$ **2 mark**

14.7. Calculate the volume of phosphoric acid (M = 97,99 g/mol) in 85% in mass and density 1,685 g/mL necessary to prepare 330 L carbonated soft drink pH 3,00. $pK_{a1} = 2{,}12$; $pK_{a2} = 7{,}21$ $pK_{a3} = 12{,}30$.

Given that pH ≈ pK_{a1}, it is true that

$[H^+] = [H_2PO_4^-] + 2 \cdot [HPO_4^{2-}] + 3 \cdot [PO_4^{3-}] + [OH^-] \approx [H_2PO_4^-]$ **1 mark**

$c(H_3PO_4) = [H_3PO_4] + [H_2PO_4^-] + [HPO_4^{2-}] + [PO_4^{3-}] \approx [H_3PO_4] + [H_2PO_4^-]$ **1 mark**

$K_{a1} = \frac{[H^+][H_2PO_4^-]}{[H_3PO_4]} = \frac{[H^+]^2}{c(H_3PO_4) - [H^+]}$ **1 mark**

$c(H_3PO_4) = \frac{[H^+]^2}{K_{a1}} + [H^+] = \frac{(10^{-3})^2}{10^{-2{,}12}} + 10^{-3} = 1{,}13 \cdot 10^{-3}\ mol \cdot L^{-1}$ **1 mark**

$c(H_3PO_4) = \frac{w_{\%} \cdot d \cdot 10}{M(H_3PO_4)} = \frac{85 \cdot 1{,}685 \cdot 10}{97{,}99} = 14{,}6\ mol \cdot L^{-1}$ **1 mark**

$V(H_3PO_4)_{conc} = \frac{c(H_3PO_4)_{dil} \cdot V(H_3PO_4)_{dil}}{c(H_3PO_4)_{conc}} = \frac{1{,}13 \cdot 10^{-3} \cdot 330}{14{,}6} = 0{,}0255\ mL$ **1 mark**

Calcium carbonate makes up different rocks and minerals. In industry it is used as a raw material to obtain quicklime (calcium oxide) by thermal decomposition.

14.8. Calculate the minimum temperature for the decomposition of calcium carbonate to be spontaneous.

$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

$\Delta H_r° = \Delta H_f°(CaO) + \Delta H_f°(CO_2, g) - \Delta H_f°(CaCO_3)$

$\Delta H_r° = -635{,}5 - 393{,}5 - (-1207) = 178\ kJ \cdot mol^{-1}$ **2 mark**

$\Delta S_r° = S°(CaO) + S°(CO_2, g) - S°(CaCO_3)$

$\Delta S_r° = 39{,}7 + 213{,}7 - 88{,}7 = 164{,}7\ J \cdot K^{-1} \cdot mol^{-1}$ **2 mark**

$\Delta G_r° = \Delta H_r° - T \cdot \Delta S_r° < 0 \quad \rightarrow \quad T > \frac{\Delta H_r°}{\Delta S_r°} = \frac{178000}{164{,}7} = 1081\ K$ **3 mark**

14.9. Explain the cause of the increase in thermal stability of the following carbonates.

Substance	$MgCO_3$	$CaCO_3$	$SrCO_3$	$BaCO_3$
Temp. decomposition	350 °C	840 °C	1010 °C	1340°C
ΔH° of decomposition	117 kJ/mol	178 kJ/mol	235 kJ/mol	267 kJ/mol

Going down in group 2, the ionic radius increases, which decreases the polarizing power of the cation, and consequently increases the thermal stability. **2 mark**

The table shows the results of the gravimetric thermal analysis of the hydrated calcium oxalate CaC_2O_4. This is a method in which the change in the mass of a sample is measured as the temperature changes. The changes in mass and the temperature at which the inflection occurs in the experimental curve are indicated in each case.

14.10. Calculate the degree of hydration of the salt.

Temperature	187 °C	504 °C	796 °C
Δm (en %)	11,86	19,01	30,19

$M(H_2O) = 2 \cdot 1{,}01 + 16{,}00 = 18{,}02\ g \cdot mol^{-1}$ **1 mark**

$M(CaC_2O_4) = 40{,}08 + 2 \cdot 12{,}01 + 4 \cdot 16{,}00 = 128{,}1\ g \cdot mol^{-1}$ **1 mark**

In 100 g of hydrate

$m(H_2O) = 11{,}86\ g \quad \rightarrow \quad n(H_2O) = \frac{m(H_2O)}{M(H_2O)} = \frac{11{,}86}{18{,}02} = 0{,}658\ mol$ **2 mark**

$m(CaC_2O_4) = 100 - 11{,}86 = 88{,}14\ g \cdot mol^{-1}$

$n(CaC_2O_4) = \frac{m(CaC_2O_4)}{M(CaC_2O_4)} = \frac{88{,}14}{128{,}1} = 0{,}688\ mol$ **2 mark**

$hydration = \frac{n(H_2O)}{n(CaC_2O_4)} = \frac{0{,}658}{0{,}688} = 0{,}956 \approx 1 \quad \rightarrow \quad CaC_2O_4 \cdot H_2O$ **1 mark**

14.11. Formulate the chemical reactions that occur if a different gas is released in each stage of decomposition.

$CaC_2O_4 \cdot H_2O(s) \rightarrow CaC_2O_4(s) + H_2O(g)$ **2 mark**

$CaC_2O_4(s) \rightarrow CaCO_3(s) + CO(g)$ **2 mark**

$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$ **2 mark**

14.12. In nature, calcium carbonate dissolves slowly by the action of acid rain, forming grottoes and caves. Calculate the solubility of calcium carbonate in a saturated solution of carbon dioxide if the total pressure is 1,00 atm. Consider the vapor pressure of water 3,17 kPa at 25 °C, the solubility product constant equal to $3{,}31 \cdot 10^{-9}$ and that **the principal equilibrium** is the following

[6] $CO_2(ac) + H_2O(l) + CaCO_3(s) \rightleftharpoons Ca^{2+}(ac) + 2HCO_3^-(ac) \quad K = ???$

The equilibrium [6] is obtained by adding the equations [3], [5] and the solubility constant, it is obtained

$K = \frac{K_{a1}' \cdot K_{ps}}{K_{a2}} = \frac{10^{-6{,}37} \cdot 3{,}31 \cdot 10^{-9}}{10^{-10{,}25}} = 2{,}51 \cdot 10^{-5}$ **3 mark**

This constant can be interpreted as a constant of the conditional solubility product.

$S = [Ca^{2+}] = 0,5 \cdot [HCO_3{}^{-}]$

$K = \frac{[Ca^{2+}]\cdot[HCO_3{}^{-}]^2}{[CO_2]} = \frac{S\cdot(2S)^2}{[CO_2]} = \frac{4\cdot S^3}{[CO_2]}$ **3 mark**

$p(CO_2) = p(total) - p(H_2O) = 101,325 - 3,17 = 98,155\ kPa$ **1 mark**

$[CO_2] = K_H \cdot p_r(CO_2) = 3,50 \cdot 10^{-2} \cdot \frac{98,155}{100} = 0,03435\ mol \cdot L^{-1}$ **2 mark**

$S = \sqrt[3]{\frac{K\cdot[CO_2]}{4}} = \sqrt[3]{\frac{2,51\cdot10^{-5}\cdot0,3435}{4}} = 6,00 \cdot 10^{-3}\ mol \cdot L^{-1}$ **1 mark**

14.13. Formalin is an aqueous solution of formaldehyde or methanal (HCHO). These solutions often contain up to 15% methanol to stabilize them as the corresponding (hemi)acetal. Justify numerically why formaldehyde is unstable from a redox point of view.

$$CO_2 \xrightarrow{-0,500} (COOH)_2 \xrightarrow{+0,102} HCOOH \xrightarrow{-0,034} HCHO \xrightarrow{+0,237} CH_3OH \xrightarrow{+0,588} CH_4$$

$\Delta E° = E°(HCHO/CH_3OH) - E°(HCOOH/HCHO) = 0,237 - (-0,034) = 0,271\ V > 0$ **3 mark**

Since the potential variation is positive, the process will be spontaneous and the formaldehyde will disproportionate.

14.14. Oxalic acid $(COOH)_2$ is used as the primary standard to standardize permanganate solutions of unknown concentration. Formulate the ionic equation of the reaction and calculate the equilibrium constant at 25°C. $E°(MnO_4{}^{-}/Mn^{2+}) = 1,507\ V$.

$2MnO_4{}^{-}(ac) + 5H_2C_2O_4(ac) + 6H^+(ac) \rightarrow 2Mn^{2+}(ac) + 10CO_2(g) + 8H_2O(l)$ **2 mark**

$\Delta E° = E°(MnO_4{}^{-}/Mn^{2+}) - E°\left(CO_2/C_2O_4{}^{2-}\right) = 1,507 - (-0,500) = 2,007\ V$ **2 mark**

$K = \exp\left(\frac{z\cdot F\cdot \Delta E°}{R\cdot T}\right) = \exp\left(\frac{10\cdot96485\cdot2,007}{8,314\cdot298,15}\right) = e^{781}$ **2 mark**

It is possible to plot the values of the free energy variation for the oxidation reaction with 1 mol of oxygen as a function of temperature for the different chemical elements. This type of graph or Ellingham diagram allows to obtain valuable information. Below is the diagram obtained for the reaction between carbon, carbon monoxide and some metals with dioxygen.

14.15. With the help of the information provided by the diagram and your knowledge of inorganics, answer or complete.

14.15.a) The most reducing metal of all those represented.

Magnesium **1 mark**

14.15.b) The metal oxide which is easiest to reduce at room temperature.

Cu_2O **1 mark**

14.15.c) The approximate temperature at which the three carbon species coexist in equilibrium.

750 °C **1 mark**

It is acceptable between 600-900 °C.

14.15.d) The most stable carbon species at elevated temperatures.

CO **1 mark**

14.15.e) In the graph, the phase changes are represented as …

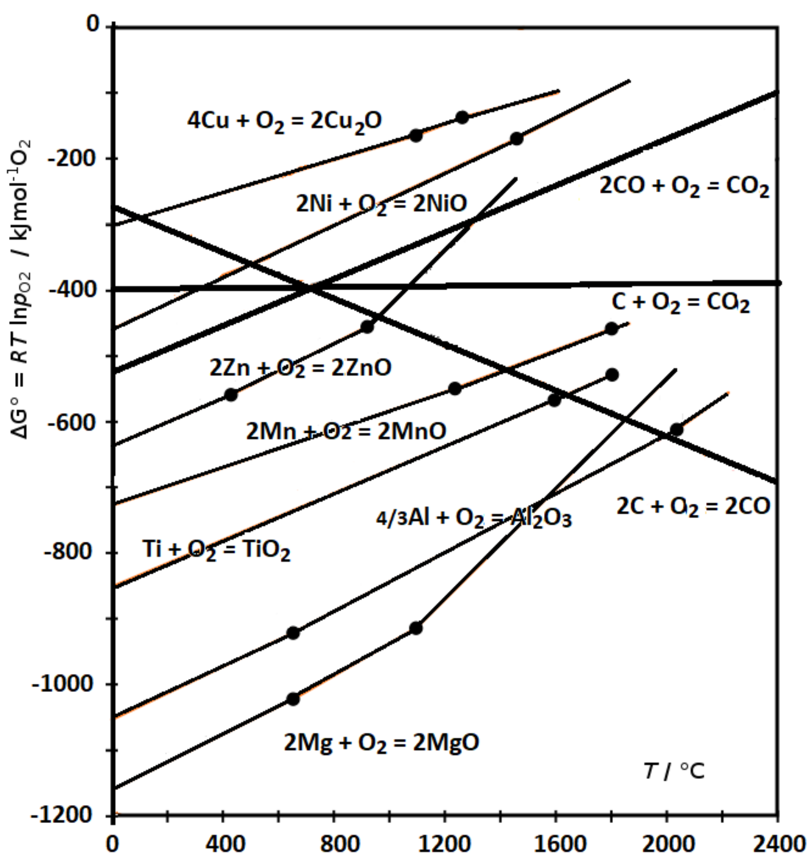
… changes in slope / points. **1 mark**

14.15.f) The temperature at which carbon can reduce zinc oxide.

950 ℃ **1 mark** It is accepted between 800-1100 ℃.

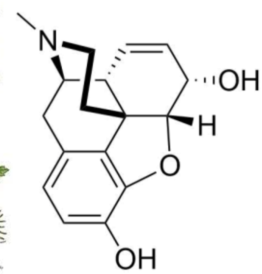
14.15.g) The free energy change changes differently for the three carbon reactions that are represented in the diagram because …

… the entropy change is different. **1 mark**

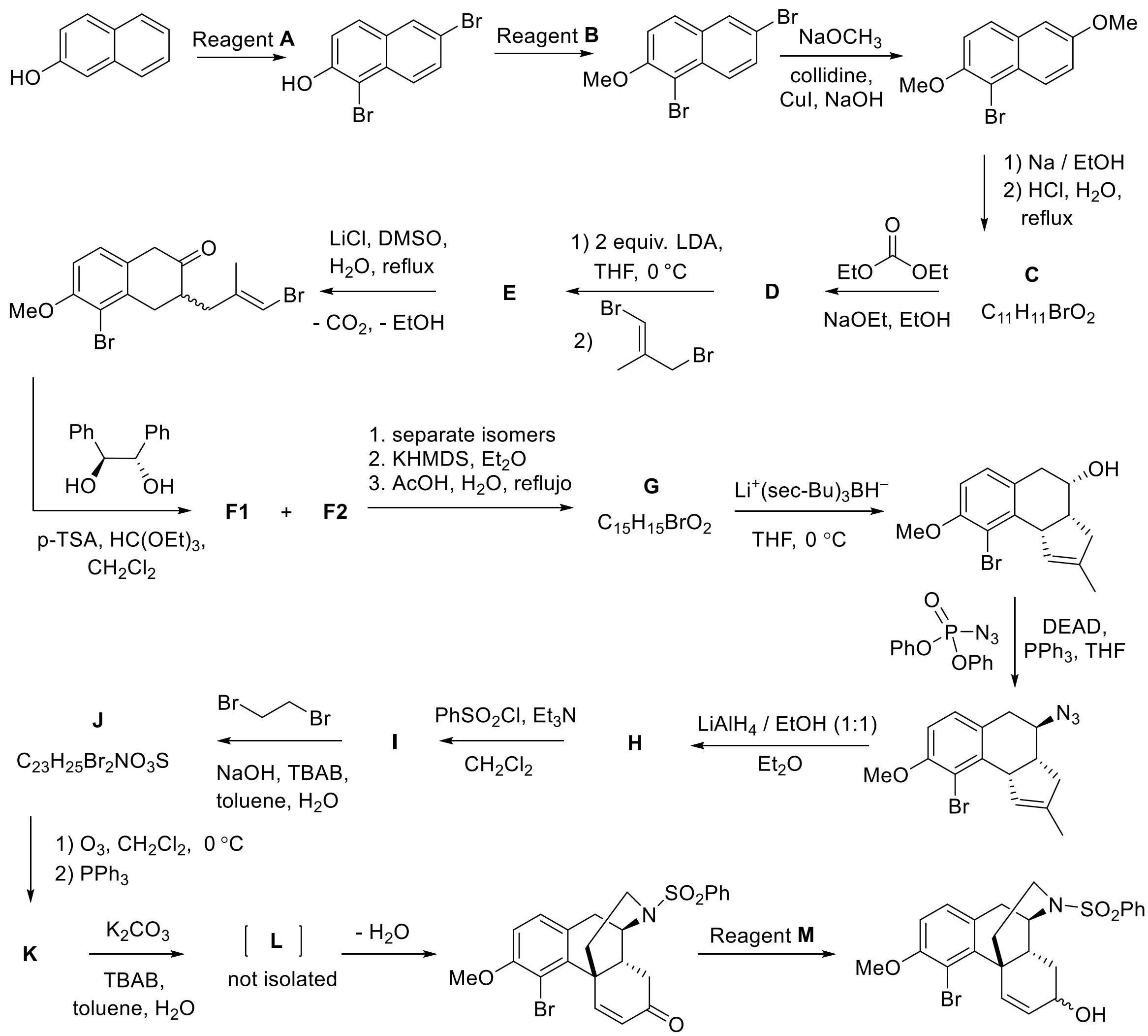


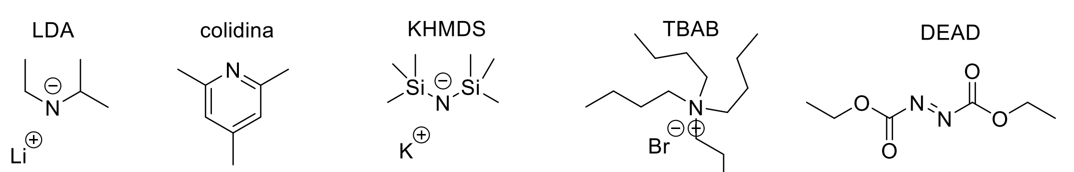
Problem 15. (-)-Morphine (12th grade)

15.1	15.2	15.3	15.4	15.5	Total	Total
20	4	8	4	4	40	10%
20	4	8	4	4	40	10%



Morphine is the main alkaloid-like component of opium, a complex mixture of substances extracted from the capsules of the opium poppy (*Papaver somniferum*). This compound has great relevance in medicine and its structure has presented a challenge for the art and science of organic synthesis. In 2002, Rheingold's group proposed a synthetic methodology for said alkaloid. (J. Am. Chem. Soc. 2002, 124(42), 12416-12417). A fragment of the synthetic sequence used is shown on the next scheme.

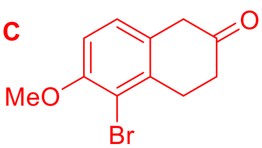




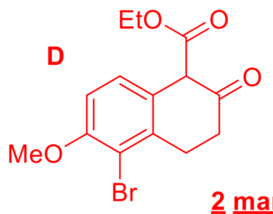
THF: tetrahydrofurane DMSO: dimethylsulfoxide DMF: dimethylformamide

Answer the following questions.

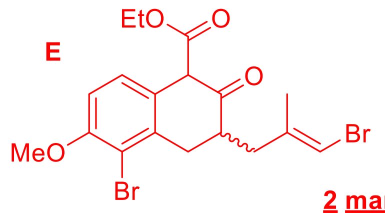
15.1. Represent the structures of compounds **C**, **D**, **E**, **F1**, **F2**, **G**, **H**, **I**, **J**, **K** and **L**.

C


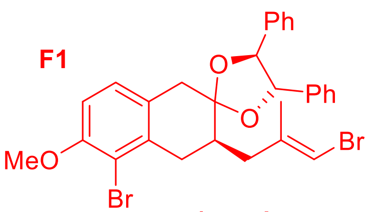
2 mark

D


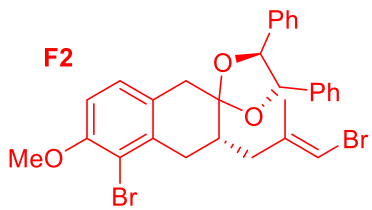
2 mark

E


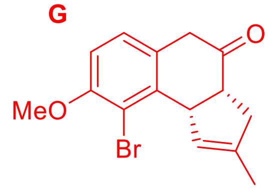
2 mark

F1


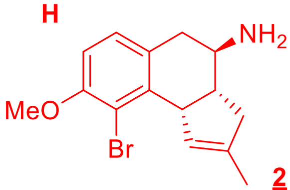
1 mark

F2


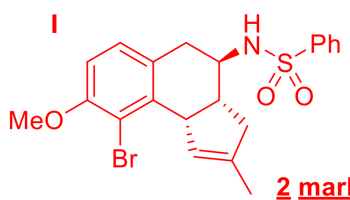
1 mark

G


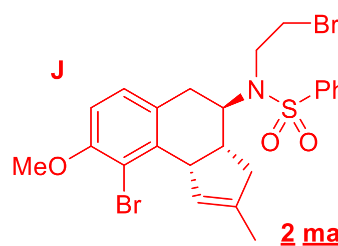
2 mark

H


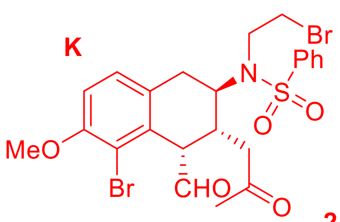
2 mark

I


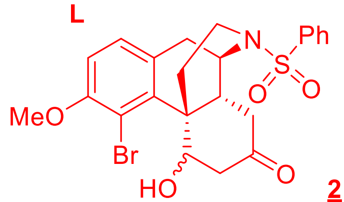
2 mark

J


2 mark

K


2 mark

L


2 mark

It is accepted to exchange F1 and F2. L can be the dicarbonyl compound, precursor of the enol.

Note: In the original syllabus, an **error** appears in the **G** formula, it appears as $C_{15}H_{16}BrO_2$, so the answers that are a consequence of this error are accepted.

15.2. Propose which (non)organic substance(s) could be used as reagents **A**, **B** and **M**.

Reagent **A**: Br_2/HOAc **1 mark**

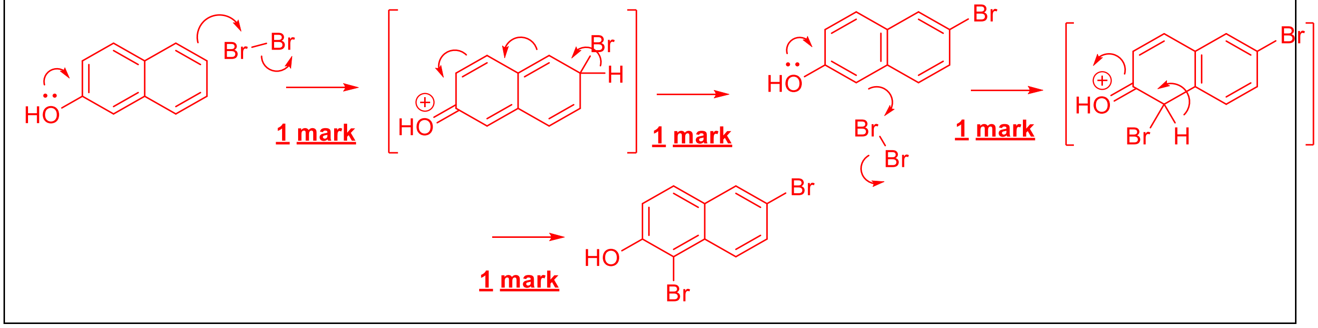
Reagent **B**: CH_3I/K_2CO_3 **1 mark**

Reagent **M**: $NaBH_4$, $CeCl_3$ **2 mark**

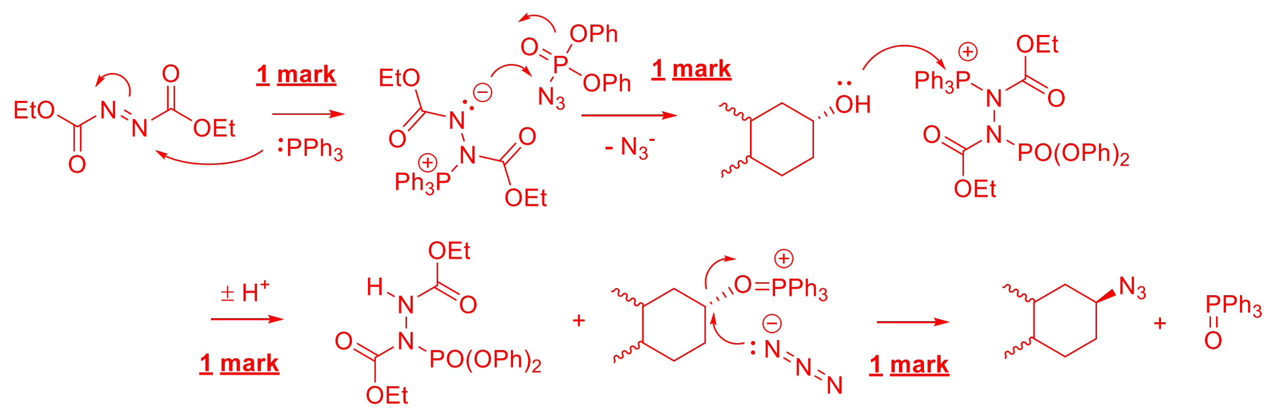
It is penalized with -1 mark if hard acid such as Ce(III) is not added.

15.3. Propose possible mechanisms for

15.3.a) the first reaction step



15.3.b) the transformation of alcohol into azide, if it is known that triphenylphosphine oxide is formed as a side product and one of the intermediates formed has two N-P bonds in its structure.



15.4. About the sequence of reactions answer:

15.4.a) What kind of isomers are **F1** and **F2**?

Diastereoisomers. **1 mark**

15.4.b) Why is the reaction with lithium tris-*sec*-butylborohydride stereoselective?

The reducing agent is a hydride that carries a very bulky ligand, so reduction occurs on the least hindered side. **1 mark**

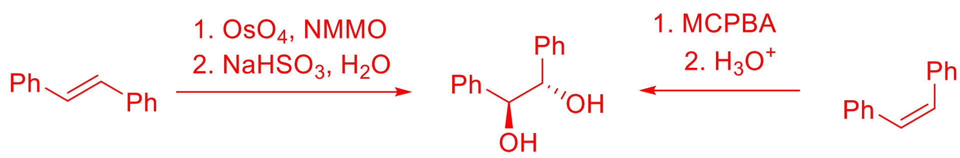
15.4.c) For what purpose is benzenesulfonyl chloride used?

Protecting group that ensures monoalkylation. **1 mark**

15.4.d) What is the function of copper(I) iodide?

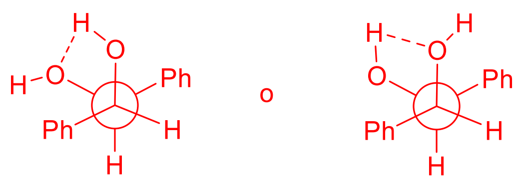
It is a catalyst that favors the rupture of the C-Br bond / the exit of Br. **1 mark**

15.5. Propose a method to obtain (1S,2S)-1,2-diphenylethane-1,2-diol (as a racemic mixture, together with its enantiomer) from the appropriate 1,2-dephenylethene isomer.



2 mark

15.5.a) Represent the Newman projection of the most stable conformation of this diol, knowing that there is an intramolecular hydrogen bond that stabilizes it.



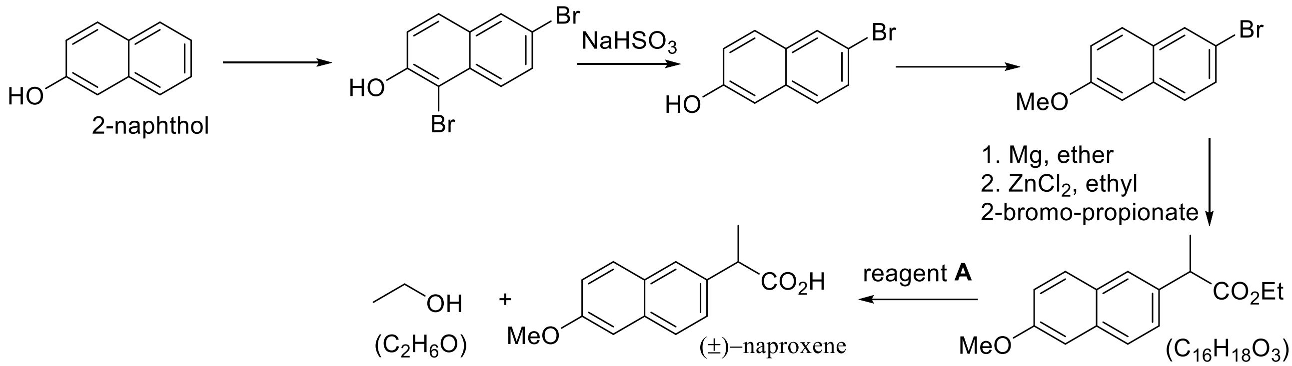
2 mark

Another conformer is acceptable if the absolute configuration is correct; this is penalized with -1 mark.

Problem 16. Aromatic compounds (12th grade)

16.1	16.2	16.3	16.4	16.5	16.6	16.7	16.8	16.9	16.10	16.11	16.12	16.13	16.14	16.15	16.16	16.17	16.18	Total	Total
10	5	11	20	1	10	2	8	2	5	4	3	6	6	6	5	10	6	120	30%
10	5	11	20	1	10	2	8	2	5	4	3	6	6	6	5	10	6	120	30%

A large part of the known organic compounds have at least one aromatic ring in their structure, so the study of their production methods and chemical properties is of great importance. Naproxen is an orally administered non-steroidal anti-inflammatory drug used to treat pain, rheumatoid arthritis, gout and fever and can be obtained according to the sequence.

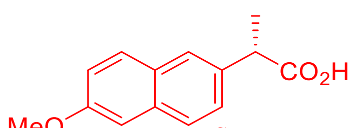


16.1. About the above sequence, please answer.

16.1.a) Naproxen is obtained in the form of a racemate, in which reaction did the racemization occur?

During the reaction of the organo-zinc compound with the ester. **2 mark**

16.1.b) Illustrate the structure of the S stereoisomer of naproxen, which is the active compound.



S-naproxene **2 mark**

16.1.c) How could you separate the S isomer from the R isomer?

By column liquid chromatography used a chiral stationary phase. / By reaction with an enantiomerically pure base and separating the diastereoisomers of the corresponding salt. **2 mark**

16.1.d) Propose a qualitative test to identify the presence of the acid group in the laboratory, if you have a sample of naproxen and additional necessary inorganic reagents.

With a mixture of KIO_3, KI, water and starch, a blue coloration is observed as a product of the formation of diiodine that consumes acid. / With K_2CO_3 and water the release of carbon dioxide is observed. **2 mark**

16.1.e) Propose possible reagents **A** that could be used to carry out the last reaction.

Reagent **A** = KOH, water. **2 mark**

The atom economy of a chemical process is a parameter used to quantify how environmentally friendly or "green" a synthesis is. It is defined as the ratio between the masses of the main product and the sum of the masses of all the reactants, assuming that the reaction is quantitative.

$$\text{atom economy} = \frac{\text{m(main product)}}{\sum \text{m(reactant)}} \cdot 100\%$$

16.2. Catalysts and solvents used in chemical reactions are not included in the atomic economy calculation.

16.2.a) Why are these substances not included in the calculation?

Because they are not part of the structure of any of the products. / Because they can be recovered. **1 mark**

16.2.b) Calculate the atom economy of the last reaction step.

$M(C_2H_6O) = 2 \cdot 12{,}01 + 6 \cdot 1{,}01 + 16{,}00 = 46{,}08\ g \cdot mol^{-1}$ **1 mark**

$M(C_{16}H_{18}O_3) = 16 \cdot 12{,}01 + 18 \cdot 1{,}01 + 3 \cdot 16{,}00 = 258{,}34\ g \cdot mol^{-1}$ **1 mark**

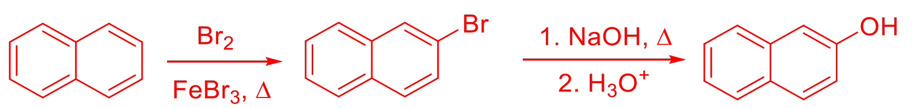
$\text{atom economy} = \frac{258{,}34-46{,}08}{258{,}34+18{,}02} \cdot 100 = 76{,}8\%$ **2 mark**

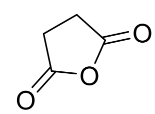
16.3. Describe possible synthetic routes to obtain the 2-naphthol used in the synthesis of naproxen from

16.3.a) Naphthalene from processed coal tar.

2 mark **2 mark**

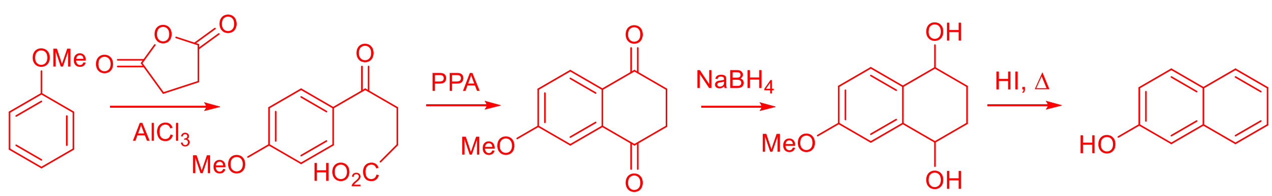
It is also accepted a synthetic route using S_NAr





succinic

16.3.b) Methoxybenzene and succinic anhydride.

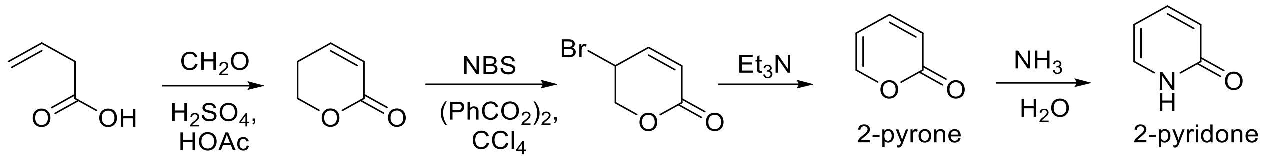


1 mark **1 mark** **2 mark** **2 mark**

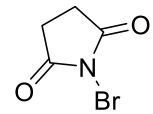
16.3.c) Which method do you consider to be more efficient?

Synthetic route **16.3.a)** is more efficient. **1 mark**

2-Pyrone is a very useful synthetic intermediate which can be obtained according to the following sequence and is converted to 2-pyridone when treated with concentrated aqueous ammonia solution.

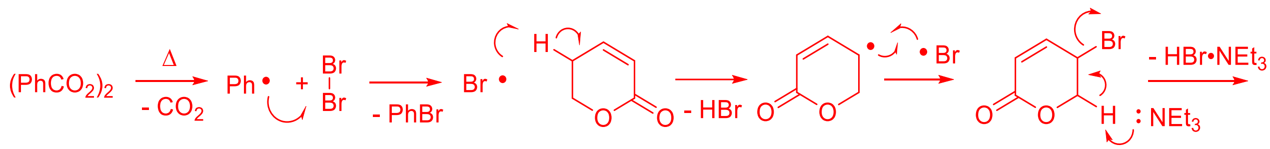


16.4. About 2-pyrone and 2-pyridone, please answer.

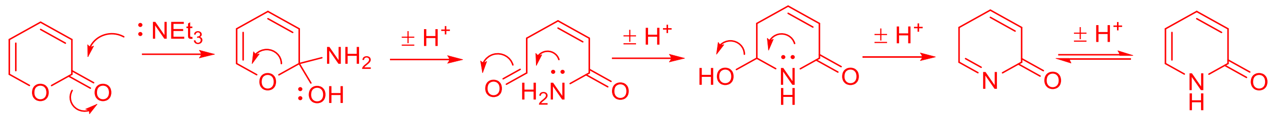


NBS: N-bromo-succinimide

16.4.a) Propose a mechanism for the last three reaction steps.



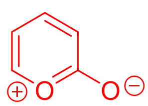
1 mark **1 mark**



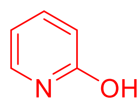
1 mark **1 mark** **1 mark** **1 mark**

16.4.b) Represent the structures of the aromatic tautomers of both compounds. Why is the non-aromatic structure the most favored in both cases?

2-pyrone
2 mark



2-pyridone
2 mark



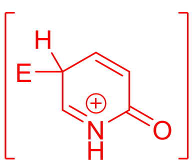
The 2-pyrone in its aromatic form generates charges. **1 mark**

2-Pyridone prefers to exist in the non-aromatic form to preserve the strong bond C=O. **1 mark**

16.4.c) 2-Pyrone can undergo electrophilic aromatic substitution reactions. What ring positions are most activated for this type of reaction? Justify your answer by representing the most relevant resonant structure of the corresponding sigma complex.

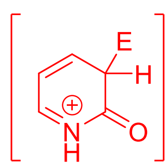
Position 5
1 mark

2 mark



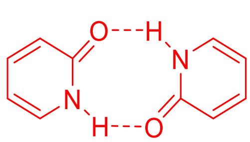
Position 3
1 mark

2 mark



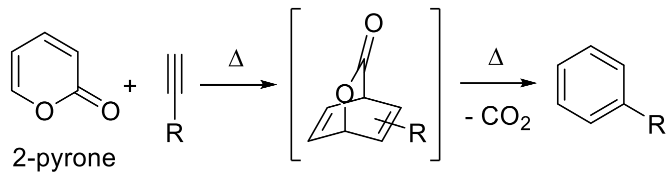
Numbering errors will not be penalized. No need to name the reactive center explicitly.

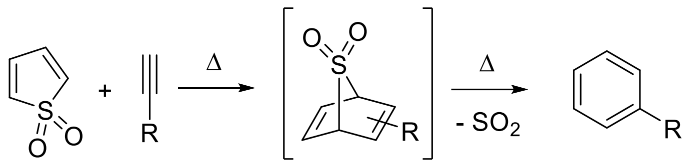
16.4.d) When 2-pyridone dissolves in nonpolar solvents it forms a dimer in which there are two hydrogen bonds. represent its structure.



2 mark

2-Pyrone, thiophene, as well as the sulfoxide and sulfone derivatives thereof are dienes and give rise to the Diels-Alder [4+2] cycloaddition reaction when reacted with suitable dienophiles. Synthetic sequences that include a cycloaddition step followed by a back-cycloaddition to remove a gas (which is favored entropically) have become standard methodology for obtaining and modifying aromatic systems, as illustrated in the following examples.

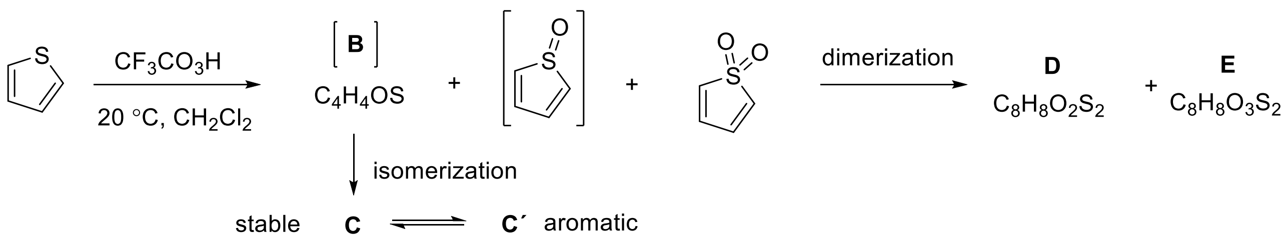
[1] 

[2] 

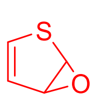
16.5. The tendency to undergo the Diels-Alder reaction is increased in the sequence thiophene < sulfoxide < sulfone. How do you explain this variation in reactivity?

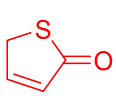
The reactivity of the dienes increases with the decrease in the aromaticity of the systems / the conjugation with the S atom. **1 mark**

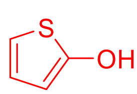
Oxidation of thiophene to the corresponding sulfoxide or sulfone can be accomplished by adding one or two equivalents, respectively, of hydrogen peroxide, oxone, or an organic sulfate peroxide. However, it is a complex process in which a series of side reactions occur, forming unwanted oxidation, isomerization and cycloaddition products.

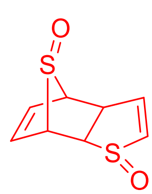


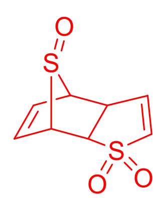
16.6. Represent the structures of **B**, **C**, **D**, **E** and the aromatic tautomer **C´**.

B  **2 mark**

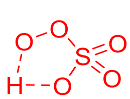
C  **2 mark**

C´  **2 mark**

D  **2 mark**

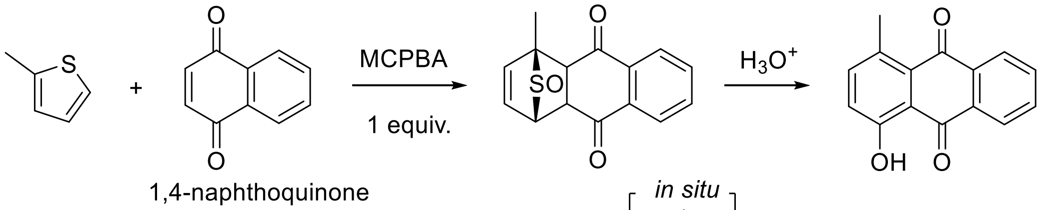
E  **2 mark**

16.7. Draw the formula for the peroxymonosulfate anion present in the triple salt Oxone $2KHSO_5 \cdot KHSO_4 \cdot K_2SO_4$ if it is known to form an intramolecular hydrogen bond.

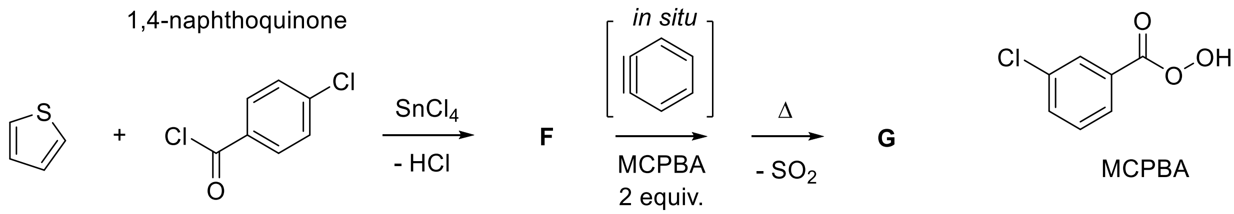
 **2 mark**

To minimize these undesirable processes, thiophene is usually oxidized *in situ*, in the presence of the dienophile with which it is going to react, as occurs in the following reaction sequences.

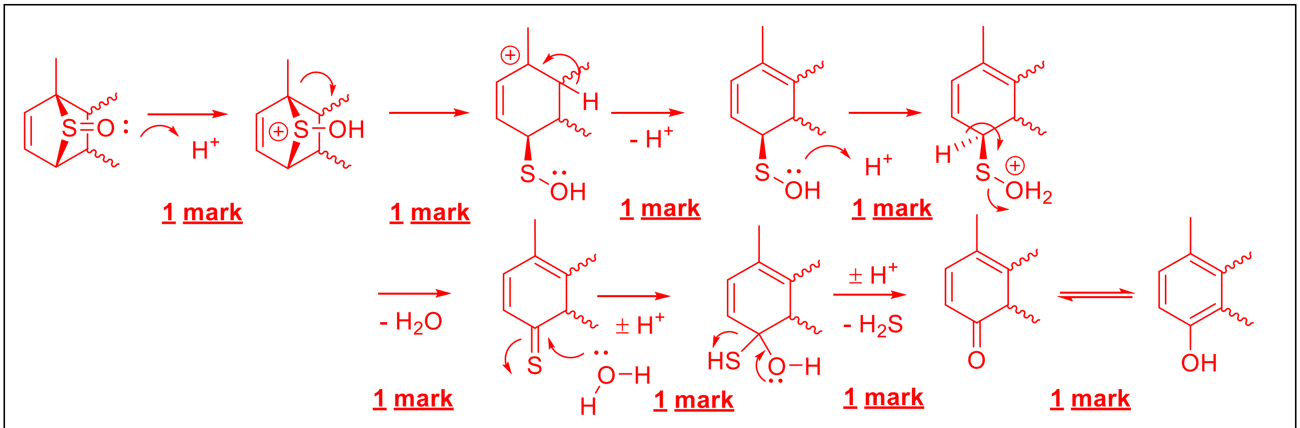
[3]



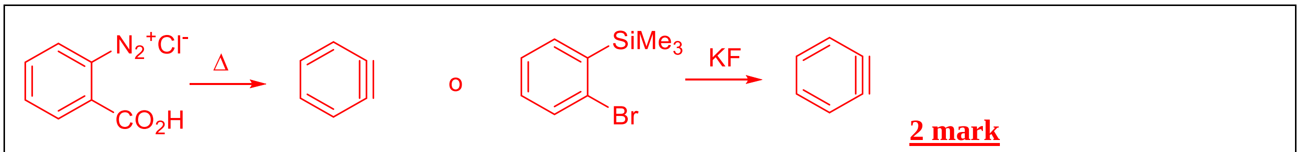
[4]



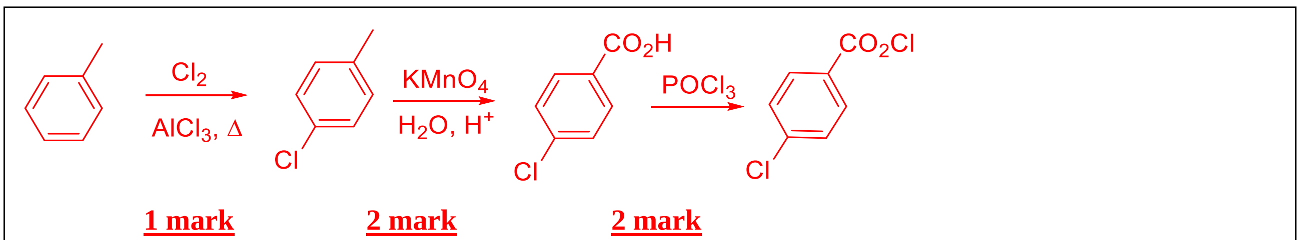
16.8. Propose a mechanism for the hydrolysis of the tricycle forming a 1,4-naphthoquinone derivative.



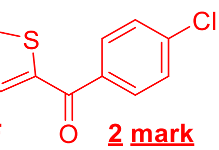
16.9. One of the reactants used is benzine. How can you get this intermediate *in situ*?



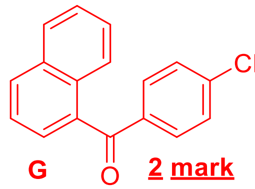
16.10. Propose a method for preparing 4-chlorobenzoyl chloride from toluene.



16.11. Represent the structures of **F** and **G**.



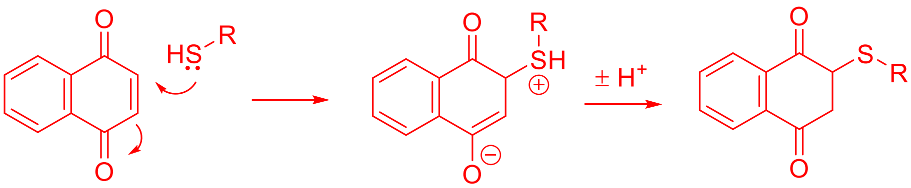
F **2 mark**



G **2 mark**

16.12. 1,4-naphthoquinone is a very toxic compound for the organism because it reacts with the thiol groups present in the glutathione tripeptide and in proteins, causing liver damage. Write down the reaction that occurs and say its name.

Michael's reaction. **1 mark**



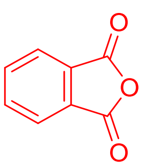
2 mark

In one of the above routes, 1,4-naphthoquinone, an oxidation product of naphthalene, is used as a starting material. Naphthalene can also be oxidized to another **H** product with the formula $C_8H_4O_3$ through the action of vanadium(V) oxide, in the so-called Gibbs-Wohl reaction. **H** has a symmetrical structure, it reacts with sodium hydroxide and amines. One molecule of **H** adds two molecules of phenol in an acid medium for the same carbon atom, with loss of water. This reaction can also be considered as two consecutive S_EAr processes. The product is the acid-base indicator phenolphthalein.

16.13. Represent the structures of **H** and phenolphthalein.

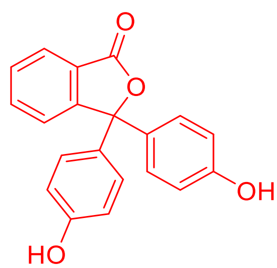
H

2 mark

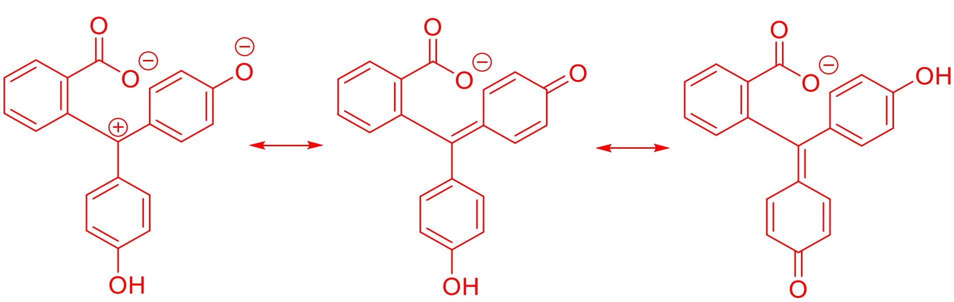


phenolphthalein

2 mark



16.13.a) Represent the structure of the anion derived from phenolphthalein if it is known that there is one less ring in its structure, compared to the neutral molecule.

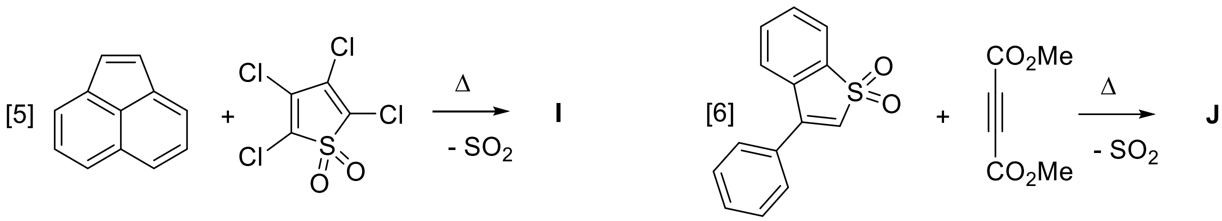


It is only necessary to represent one structure. **2 mark**

The Diels-Alder reaction can be classified as normal electron demand (NED) when the diene donates electrons to the dienophile, reverse electron demand (IED) when the diene accepts electrons from the dienophile, or intermediate type when there is virtually no charge transfer.

16.14. For the following reactions

16.14.a) Classify them as NED, IED or intermediate type cycloadditions.

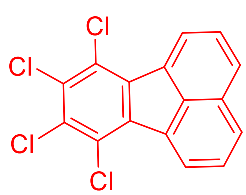


Reaction [5] is predicted as inverse electron demand (IED). **1 mark**

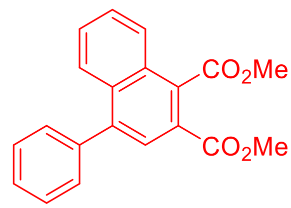
The reaction [6] is predicted as normal electron demand (NED). **1 mark**

16.14.b) Represent the structures of the aromatic products **I** and **J**.

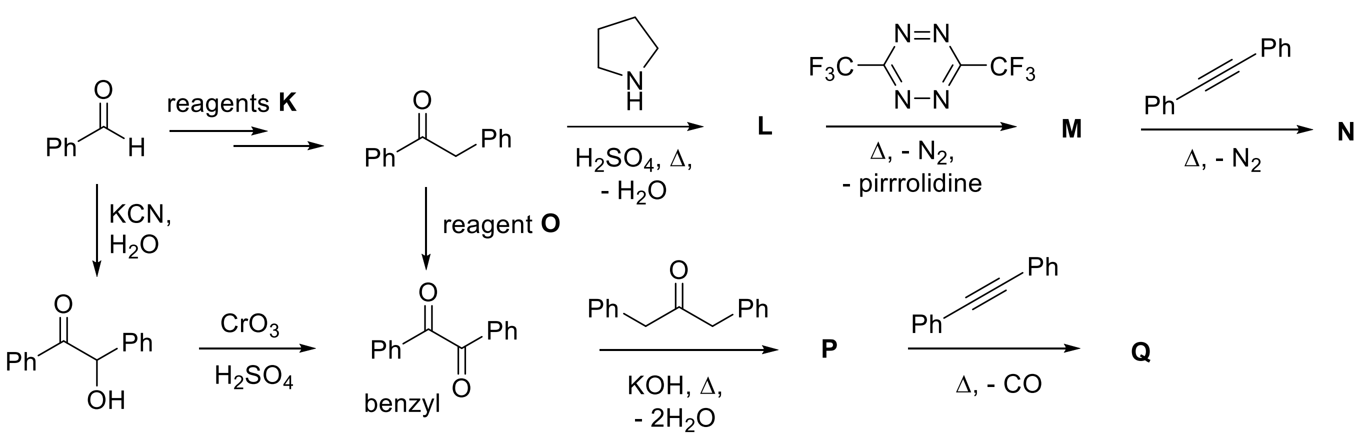
I **2 mark**



J **2 mark**



The following synthetic routes allow the formation of aromatic compounds with high symmetry. All use one or more (retro)-Diels-Alder reactions.



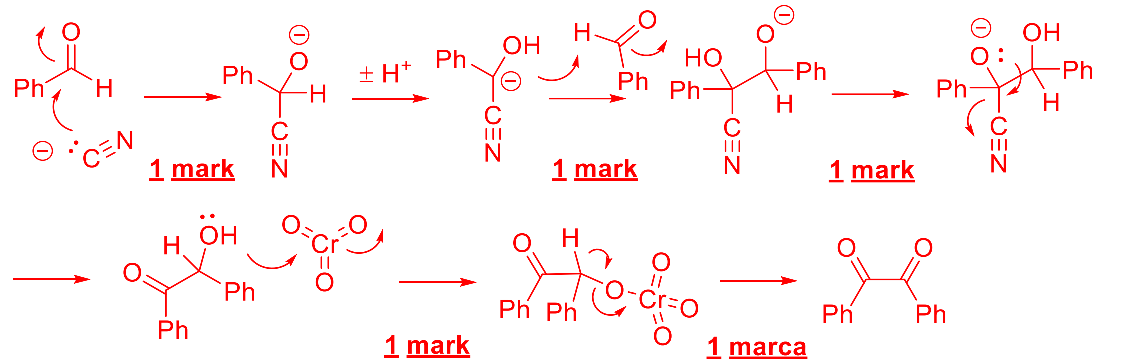
16.15. Propose reagents or a mixture of reagents **K** and **O** to carry out the given transformations.

Reagent **K** = 1. TiO_2, 2. H_3O^+, 3. CrO_3 o 1. $PhCH_2MgBr$, 2. H_3O^+, 3. CrO_3 **4 mark**

Reagent **O** = SeO_2 o 1.LDA, 2. Davis oxaziridine, CrO_3 o 1. Me_3SiCl, imidazole, 2. mCPBA, 3. H_2O

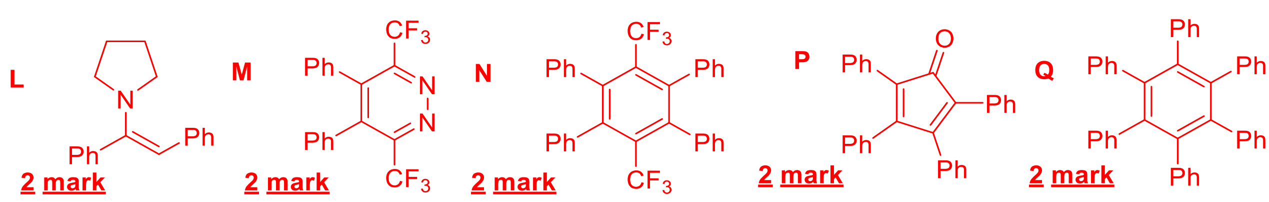
2 mark

16.16. Propose reaction mechanisms for the two steps of transformation of benzaldehyde into benzyl diketone.

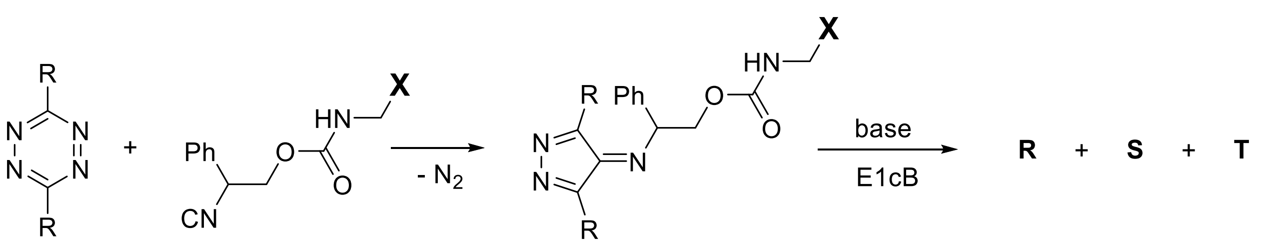


It is acceptable a mechanism based on the reactions used in **16.15.**

16.17. Represent the structures of **L**, **M**, **N**, **P** and **Q**.



The 2022 Nobel Prize in Chemistry was awarded to the Dane Morten Meldal and the Americans Carolyn Bertozzi and Barry Sharpless, who already had a previous award, for "the development of Click chemistry and bioorthogonal chemistry", according to the jury of the Royal Swedish Academy of Sciences in its decision. A reaction that recently began to be used for applications of this type is the cycloaddition [4 + 1] between 1,2,4,5-tetrazines and isonitriles. The following scheme illustrates the use of this reaction for the activation of a protein **X** that contains an amino group of a lysine amino acid that is temporarily blocked in the carbamate form. (ACS Omega 2020, 5, 40, 25505-25510).



16.18. Propose structures for the products **R**, **S** and **T**.